

Sending mixed signals: convergent iridescence and divergent chemical signals in sympatric sister-species of Amazonian butterflies

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eLife Assessment

This study presents a **valuable** assessment of and **solid** evidence for increased similarity in visual appearance combined with increased chemical differences between two butterfly species in sympatry compared with differences between three populations of one of the two species in allopatry. The similarity in visual appearance hints to an evolutionary response to shared predators (but alternative explanations are possible). Thus, the difference in chemical signaling likely helps to avoid between-species mating in sympatry.

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Abstract

The evolution of traits in closely related species living in sympatry strongly depends on both shared selective pressures and reproductive interference. In closely-related *Morpho* butterfly species living in the understory of the neo-tropical rainforest, the blue iridescent coloration of the wings is likely involved in predation evasion, as well as in mate recognition and courtship. We used spectrophotometry, behavioral experiments, visual modeling and chemical analyses to characterize the evolution of visual and chemical traits in two closely-related species, *Morpho helenor* and *Morpho achilles*. We specifically compared trait variation between samples from allopatric and sympatric populations of *M. helenor* to test the effect of ecological interactions with *M. achilles* on trait evolution. We quantified the differences in wing iridescence and tested for variations in the sexual preference for this trait. We found a strong similarity in iridescence between *M. helenor* and *M. achilles* in sympatry while the iridescence of *M. helenor* diverged in allopatry, suggesting that predation favors local resemblance. Although intraspecific behavioral experiments showed that iridescent signals

could be used as visual cues during intraspecific mate choice, the strong resemblance of the iridescent signals between species may impair these species visual recognition. In contrast, the divergent chemical bouquets detected between species suggest that the visual similarity of sympatric *Morpho* species may have favored the divergence of alternative traits involved in species recognition, such as chemical cues.

Introduction

Numerous ecological interactions can impact trait evolution in closely-related species in sympatry, leading to trait convergence or divergence (terHorst et al., 2018 [DOI](#)). Closely-related species often display similar suites of traits because of their shared evolutionary history (Blomberg et al., 2003 [DOI](#)). Local selection can also act as a filter and prevent trait divergence (Keddy, 1992 [DOI](#)), therefore enhancing trait similarity between closely-related species when they occur in sympatry (Chazot et al., 2014 [DOI](#)). Trait similarity among sympatric species within a given ecological niche can thus stem from retention of locally adapted ancestral traits or from evolutionary convergence (Muschick et al., 2012 [DOI](#)). Selective pressures promoting the retention of locally adapted traits within species and/or trait convergence among sympatric species can also be due to local ecological interactions: for instance, shared predation pressure may promote the convergence of predator-deterrent traits in sympatry, but allow the trait to differentiate in allopatry (Mallet, 1999 [DOI](#)).

However, when closely-related sympatric species share a given trait, either as a result of ancestry and/or convergence, they often diverge in other traits because (1) they may be partitioned in different ecological niches (Berlocher & Feder, 2002 [DOI](#)), or (2) as a result of character displacement due to reproductive interference (Grether et al., 2020 [DOI](#)) or reinforcement due to poor hybrid fitness (Butlin & Smadja, 2018 [DOI](#)). As a result, traits involved in sexual competition or mate choice tend to diverge significantly more often between species in sympatry compared to allopatry (Haavie et al., 2004 [DOI](#); Marko, 2005 [DOI](#)).

In this study, we investigate how ecological interactions in sympatry can constrain natural and sexual selection shaping trait evolution. We specifically focus on traits submitted to both natural and sexual selection, and compare differences in these traits in allopatric vs. sympatric ranges.

Theoretical and empirical studies have shown that sexual selection may favor the evolution of preferences for locally-adapted traits within species (Servedio, 2004 [DOI](#); Servedio & Boughman, 2017 [DOI](#); van Doorn et al., 2009 [DOI](#)). For instance, the predator-deterrent coloration of poison frogs is also detected and used as mating cues by females (Reynolds & Fitzpatrick, 2007 [DOI](#)). Similarly, habitat-dependent coloration of sympatric cichlid fish is also used as a visual cue for mate recognition (Seehausen et al., 2008 [DOI](#)). Yet, sexual interactions are likely to occur between individuals from closely-related species when they live in sympatry, and similar preferences for adaptive traits may thus result in substantial reproductive interference (Gröning & Hochkirch, 2008 [DOI](#); Soni et al., 2024 [DOI](#)). Hybrids, when produced, can be unfit, thus favoring the evolution of sexual preferences for species-specific cues, rather than locally-adapted traits (Maisonneuve et al., 2024 [DOI](#)). To determine to what extent ecological interactions shape trait evolution, it is thus necessary to compare patterns of trait evolution in sympatry and allopatry: allopatric populations indeed allow to estimate background levels of divergence or similarity that arise in the absence of direct ecological interactions (Pfennig & Pfennig, 2009 [DOI](#)). Comparing variations in adaptive traits in sympatric vs. allopatric populations of recently-diverged species, and testing the sexual preference for those traits can shed light on the selective processes targeting traits modulating reproductive isolation and co-existence in sympatry.

In butterflies, the evolution of wing color patterns can be influenced by both natural and sexual selection. The visual discrimination of wing color pattern can enable intraspecific recognition during courtship in many species (Costanzo & Monteiro, 2007 [↗](#); Li et al., 2017 [↗](#)). However, the evolution of wing color patterns is also strongly influenced by the risk of detection and/or recognition by predators (Finkbeiner et al., 2014 [↗](#); Oliver et al., 2009 [↗](#)). Whether these opposite selective pressures ultimately promote trait convergence or divergence in sympatric species might depend on their relatedness: for instance, a recent study in *Papilionidae* showed multiple color pattern convergences between distantly-related species living in sympatry, while divergent colorations are found in closely-related species (Puissant et al., 2023 [↗](#)). Divergence in traits involved in species recognition could be favored because of higher reproductive interference in closely related species than in distantly related taxa (Pfennig & Pfennig, 2009 [↗](#)). In sympatric species with chemical-defenses, such as *Heliconiini* butterflies, local predation pressures tend to promote the convergence of similar conspicuous warning wing patterns compared to allopatric species (*i.e.* Müllerian mimicry, Joron et al., 1999 [↗](#); Merrill et al., 2014 [↗](#)). But the costs associated with hybrid production in turn favor the evolution of alternative divergent mating cues in mimetic butterflies (Estrada & Jiggins, 2008 [↗](#)), and divergence in male pheromone bouquets and female attraction has been found among mimetic sister species (González-Rojas et al., 2020 [↗](#)). Similarly, the evolution of specific visual mate recognition signals, limiting reproductive interference but indistinguishable by predators, can also be promoted on the wings of mimetic butterflies (Laurens et al., 2014 [↗](#)).

Here we focus on the evolution of mating cues in the neotropical butterfly genus *Morpho*, where multiple closely-related species co-exist in sympatry (Blandin & Purser, 2013 [↗](#)). In the *Morpho* species observed in the understory, striking iridescent blue coloration is displayed on the dorsal side of the wings, due to specific wing scale structure (Giraldo et al., 2016 [↗](#); Siddique et al., 2013 [↗](#)). The light signal reflected by iridescent surfaces can be very directional, as hue and brightness of iridescent objects can drastically change depending on the light environment or the observer's position (Doucet & Meadows, 2009 [↗](#)). While the iridescent blue color is probably ancestral to the diversification of the understory clade (Chazot et al., 2021 [↗](#)), the precise reflectance spectra at different angle likely differs among *Morpho* species. Directional iridescent signals generated in animals can likely enhance recognition by mates while remaining poorly detected by predators (Endler, 1992 [↗](#)). In birds (Simpson & McGraw, 2019 [↗](#)) and butterflies (White et al., 2015 [↗](#)), the specific directional signal produced by the iridescent trait can be used as a cue during courtship, suggesting that the antagonistic sexual and natural selective pressures may finely tune the evolution of iridescent effects. How much sexual selection shapes the evolution of iridescent properties in sympatric *Morpho* species is currently unknown, but behavioral experiments carried out in the field in Amazonian Peru highlighted strong visually-based territorial interactions among males from sympatric species and limited species discrimination based on female coloration in males (Le Roy, Roux, et al., 2021).

This raises questions on the key visual cues involved in mate choice, given that the iridescent blue coloration shared by closely-related species encountered within the understory is also likely under selection by predators. The iridescent bright blue dorsal coloration of *Morpho* wings contrasts with the brown and matte ventral side, and generates a peculiar visual effect during flight. The combination of the blue flashes produced by the alternate exposure of the bright blue vs. brown sides of the wings during flapping flight, in addition to erratic flight trajectories, makes these *Morpho* very difficult to catch by bird predators (Young, 1971 [↗](#)), potentially enhancing their evasive capabilities (Murali & Kodandaramaiah, 2020 [↗](#)). Experimental trials with evasive prey have shown that predators learn to avoid prey they repeatedly fail to catch (Páez et al., 2021 [↗](#)). The display of iridescent wings could thus be associated with a higher survival rate in nature because of both (i) direct effects, through successful escape of predator attacks, and (ii) indirect effects, by limiting predation attempts by birds recognizing the blue signals and refraining from attacking, as highlighted by butterfly release experiments investigating the hunting behavior of wild insectivorous birds in Brazil (Pinheiro et al., 2016 [↗](#)). Mark recapture experiments in the field

with manipulated dorsoventral contrasts in wild *Morphos* have suggested that dynamic flash coloration can reduce predation rate (Vieira-Silva et al., 2024 [↗](#)). This indirect effect could promote the evolution of convergent blue patterns in sympatric species, similar to the mimicry observed in species with chemical-defenses (Joron et al., 1999 [↗](#); Merrill et al., 2014 [↗](#)). In line with this hypothesis, repeated local convergence in the proportion of iridescent blue vs. black areas on the dorsal side of the wings has been documented in the sister-species *Morpho helenor* and *Morpho achilles* living in sympatry throughout the Amazonian basin (Llaurens et al., 2021 [↗](#)). Precise quantification of variations in iridescence is now needed to assess the respective effects of selection by predators and mates, that may drive convergent vs. divergent evolution of iridescence in sympatric and allopatric ranges.

First, we quantified iridescence in allopatric vs. sympatric populations of *M. helenor* subspecies. Since the coloration is expected to be more similar within than among species under neutral evolution, we used allopatric populations of *M. helenor* as a baseline to assess convergence of iridescence between two sympatric species (*M. helenor* and *M. achilles*). We then conducted behavioral experiments to test the effect of variation in iridescence on mate recognition, using two subspecies of *M. helenor* displaying different iridescent phenotypes. This intraspecific comparison allows identifying the visual cues used in mate choice in *M. helenor*, teasing apart the effects of iridescence and/or wing pattern. We then tested whether those visual cues are used in species recognition between sympatric *M. helenor* and *M. achilles*. Finally, we studied variations in the volatile compounds produced by wild males and females from sympatric populations of the two species to explore the evolution of potentially alternative traits, such as chemical cues, possibly acting as a reproductive barrier.

Results

We investigated differences in wing iridescence, color perception and color pattern-based mate choice of two sister species of *Morpho*, *M. helenor* and *M. achilles*, which diverged ~ 3.6 Mya (Chazot et al., 2021 [↗](#)). We relied on geographic variations of wing color patterns within *M. helenor* (Fig 1 [↗](#)) to compare the iridescence of three different populations of *M. helenor* (subspecies *M. helenor bristowi*, *M. helenor theodorus* and *M. helenor helenor*) and test the perception of these patterns by predators and mates using visual models.

The convergent iridescent signals of sympatric *Morpho* species are not differentiable

We measured the reflectance of *Morpho* wings at different angles of observation and illumination, in two different planes (referred to as proximo-distal and antero-posterior planes) following two complementary protocols (referred to as “specular” and “tilt”; see Fig 2.A [↗](#) and Supporting Information Text 1 for extended methods). We then extracted for each spectrum three commonly-used colorimetric variables: Hue, Brightness and Chroma (analyzed in S4 Fig).

We found that hue and brightness always vary depending on the angle of incidence of the light, as confirmed by PERMANOVAs detecting a significant effect of the “Angle” for all methods of measurements (Fig 2 [↗](#) and statistical analyzes in S1 Table and S5 Fig) confirming that in both species, both sexes are iridescent. Second, the brightness of *Morpho* wings is highly sexually dimorphic, as the effect of sex was always significant. The brightness measured with the “tilt” protocol (Fig 2.F [↗](#) and 2.G [↗](#)) showed that males tend to be brighter than females in all populations.

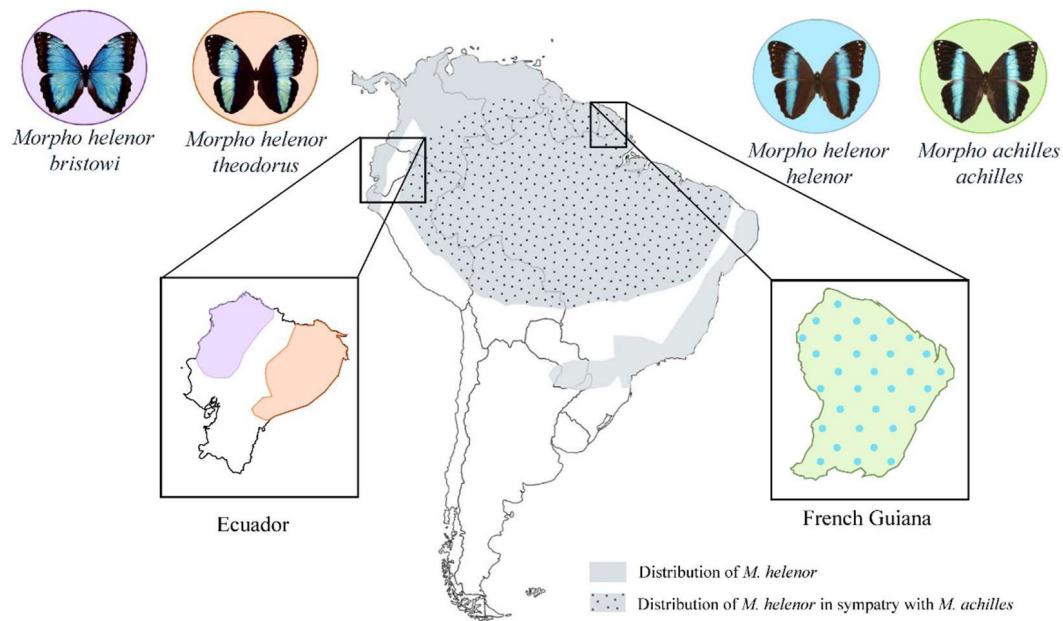


Fig 1

Distribution of the sister-species *M. helenor* and *M. achilles* across South America.

The gray area represents the whole distribution of *M. helenor*, and the dotted area indicates the localities where *M. helenor* is in sympatry with *M. achilles*. Note that the different subspecies of *M. helenor* are found in different localities and display substantial variations in the proportion of black vs. blue areas on the dorsal sides of their wings. For example, *M. h. bristowi* and *M. h. theodorus* are allopatric in Ecuador, with their respective distribution separated by the Andes such that they are never in contact. While *M. h. bristowi* is found on the Pacific side of the country and has a wide blue band on the dorsal side of its wings, *M. h. theodorus* is found in Western Amazonia and displays a narrow blue band on the dorsal side of its wings. However, *M. helenor* is sympatric with *M. achilles* throughout the Amazonian rainforest. In French Guiana, the subspecies *M. helenor helenor* and *M. achilles achilles* display convergent dorsal color patterns with thin blue bands (Llaurens et al., 2021 [\[1\]](#)).

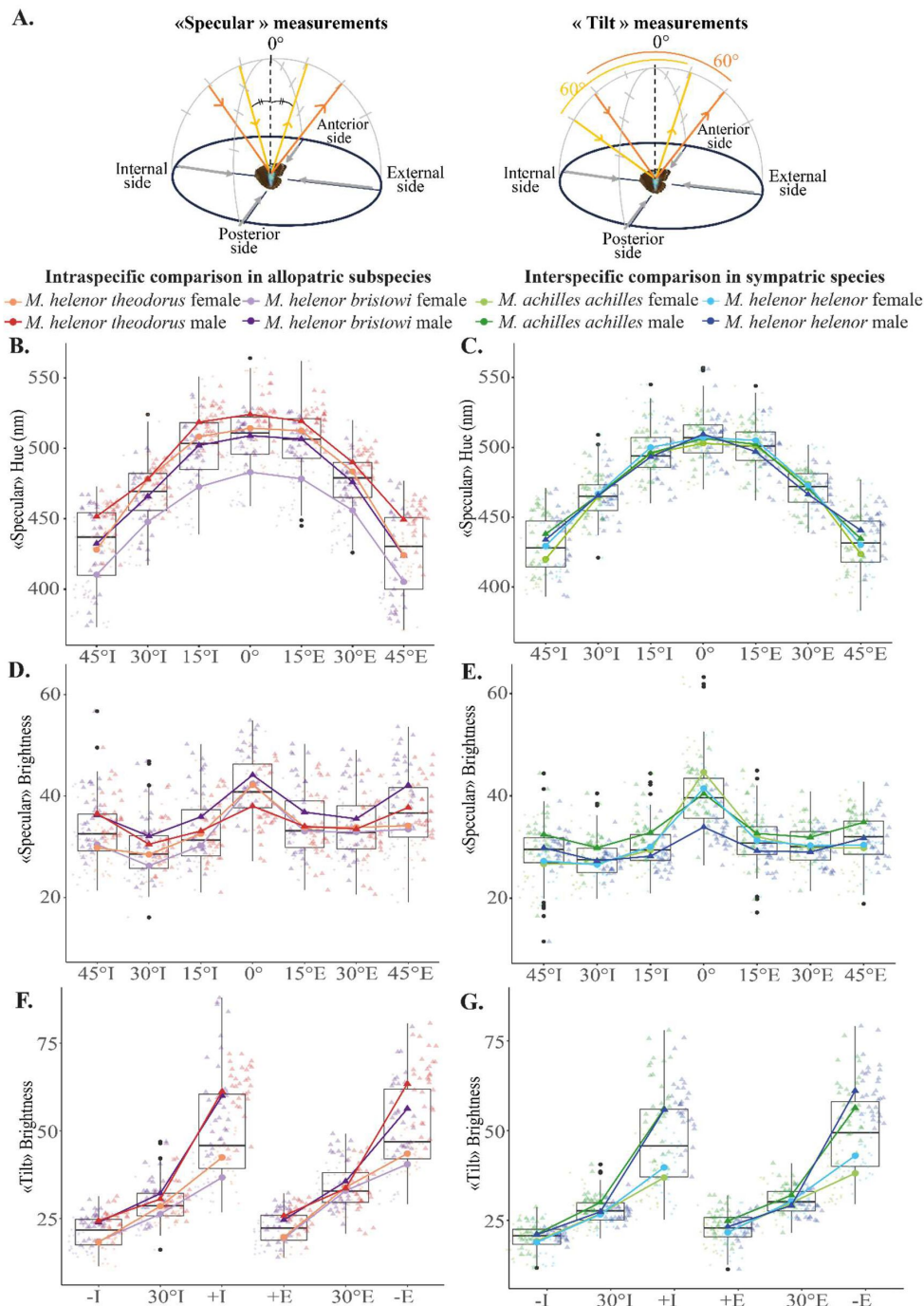


Fig 2

Differences of hue and brightness on the proximo-distal plane of *Morpho* wings

(A) Illustration of the two protocols used to assess the differences in *Morpho* wing reflectance. The “Specular” set-up allows for the quantification of the wing color variations, while the “tilt” set-up can be used to quantify brightness variation for each *Morpho* wing (see Appendix for extended methods). Variations of hue (B and C) and brightness (D and E) calculated from the wing reflectance measured with the “Specular” set-up, and variations of brightness calculated from the wing reflectance measured from the “tilt” set-up (F and G). Those hue and brightness parameters were calculated for the allopatric *M. h. theodorus* and *M. h. bristowi* (first column in orange and purple) and for the sympatric *M. h. helenor* and *M. a. achilles* (second column in green and blue) on the proximal-plane plane of their wings (I= illumination on the internal side of the wings, E= illumination on the external side of the wings). See S1 Table for the PERMANOVA analyses describing those graphs.

To finely appraise the complete spectrum of variation in iridescence within and among *Morpho* species, we then used a multivariate approach allowing to analyze the full reflectance spectra obtained from both proximo-distal and anteroposterior measurements. At the intraspecific level, significant divergence of iridescence was detected between allopatric subspecies of *M. helenor* (Fig 3.A). In contrast, iridescence did not differ between species living in sympatry (PERMANOVA, $F=2.09$, $df=1$, $p\text{-value} = 0.084$), suggesting that iridescence converges in sympatry (Fig 3.B). In both cases, the interaction term *taxon***sex* was non-significant, suggesting that the sexual dimorphism of iridescence was similar for all taxa.

We then modeled the perception of color contrast in the visual systems of *M. helenor* and UV-sensitive birds respectively, and tested whether the differences of reflectance observed between sexes and species at different angles of illumination could potentially be perceived by a *Morpho* observer or a putative predator. As sexual dimorphism of wing reflectance was detected, we measured the achromatic distances and chromatic distances for males and females separately. Achromatic distances results (S6 Fig) suggest that neither birds nor *Morpho* butterflies could theoretically differentiate the brightness of the wings of sympatric or allopatric individuals, based on visual modelling. Concerning chromatic distances, as the results were similar along the two directions of measurements (antero-posterior and proximo-distal), only proximo-distal results are shown in Fig 3.C (antero-posterior results in S7 Fig). Results suggest that *Morpho* butterflies and predators can theoretically visually perceive the difference in the blue coloration between the different subspecies of *M. helenor*: mean chromatic contrasts were always higher than 1 JND, for both sexes and for all angles, using both bird and *Morpho* visual models. However, this effect was stronger for females. In males, confidence intervals were only strictly significant for the 0° illumination using the *Morpho* visual model, suggesting a weak discrimination at most angles of observation.

In contrast, the mean chromatic contrasts measured between sympatric species were always below 1 JND: the minute differences observed between sympatric species are thus unlikely to be discriminated either by *Morpho* or bird predators.

Evidence of mate preferences based on visual cues within species

Experimental evidence has shown that *M. helenor* males are attracted to blue coloration when patrolling in the wild (Le Roy, Roux, et al., 2021). We thus performed a series of mate choice experiments to determine the nature of the visual cues used in mate choice in *M. helenor*. We focused on the two previously studied allopatric subspecies of *M. helenor* (*M. h. bristowi* and *M. h. theodorus*), which significantly differ in both wing pattern and iridescence.

We first tested whether mating tended to be preferentially assortative by conducting a tetrad experiment involving one male and one female of both subspecies. The highest number of mating events occurred between the two assortative pairs (see Table 1). Indeed, a Fisher's exact test performed on the contingency table showed a significant departure from random mating ($p\text{-value} = 0.04$). The time until an assortative mating ($mean = 48$ min, $sd = 38$ min) or until a disassortative mating ($mean = 48$ min, $sd = 38$ min) however did not significantly differ ($T\text{-test}$; $stat = -0.27$, $p\text{-value} = 0.79$). Note that the effects were clearly driven by the bolder behavior of *M. h. bristowi* individuals, as half of the mating events occurred between a female and a male of this subspecies.

To test whether this assortative mating was driven by visual cues and to identify these cues, we performed a series of experiments using dummy butterflies treated with hexane to remove any olfactory cue. We specifically focused on studying allopatric *M. h. theodorus* and *M. h. bristowi* because they both differ in their iridescent signal and in their iridescent pattern. Different pairs of female dummies were presented to *M. helenor* males to investigate assortative preference between (sub)species (experiment 1), discrimination of the wing pattern (experiment 2) and discrimination of the iridescent coloration (experiment 3) (see the experimental details in the M&M section).

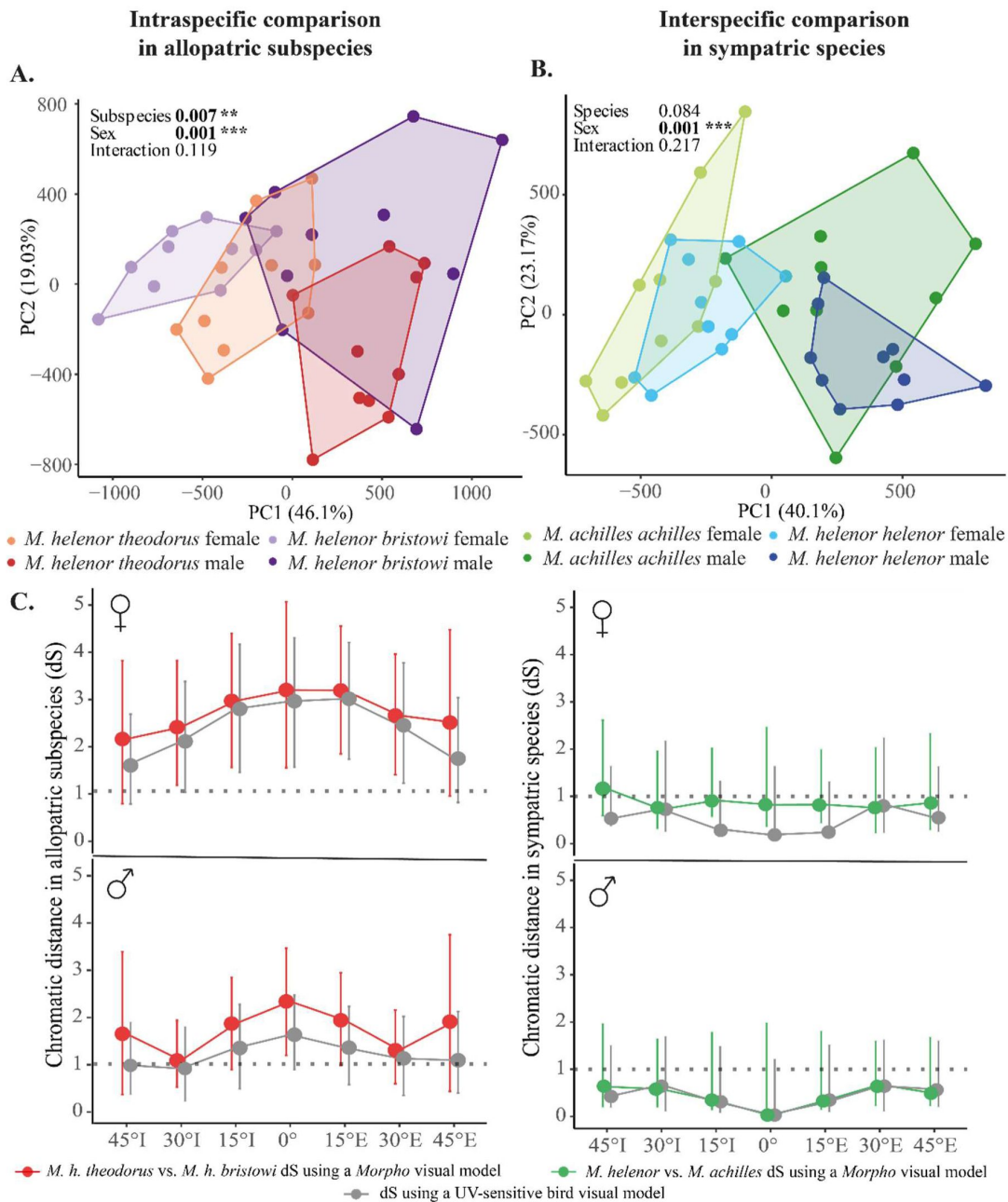


Fig 3

Characterization and perception of the iridescent coloration of *Morpho* butterflies.

(A) and (B): PCAs showing the variation in iridescence for both sexes, (A) in the two sympatric species from French Guiana (*M. h. helenor* vs. *M. a. achilles*) and (B) in the two allopatric Ecuadorian subspecies of *M. helenor* (*M. h. theodorus* vs. *M. h. bristowi*). Each point represents the global signal of iridescence of each individual, corresponding to the 21 complete reflectance spectra obtained from the 21 tested angles of illumination. The results of the PERMANOVA are shown on the top left corner of each graph. (C) The chromatic distances (*i.e.* the visual discrimination rate by a visual model) of the wing reflectance measured with the “Specular” set-up on the proximo-distal plane. Visual modeling was used to calculate the chromatic contrast of blue coloration between allopatric *M. helenor* subspecies (red) and between the two sympatric sister-species *M. helenor* and *M. achilles* (green), as perceived by a *Morpho* visual system for every angle of illumination measured on the proximo-distal plane. The chromatic contrast likely perceived by UV-sensitive birds is shown in gray. Chromatic contrast of the female wings (top) and male wings (bottom). The threshold of discrimination is shown by the dotted line and set to 1 *Just Noticeable Difference* (JND). Error bars show the confidence intervals calculated during the bootstrap analysis.

N=30

	M. h. theodorus female	M. h. bristowi female
M. h. theodorus male	6	3
M. h. bristowi male	5	16

Table 1

Tetrad experiment results:

Number of mating events involving the different possible pairs of males and females from the two subspecies of *M. helenor* (bristowi vs. theodorus) in 30 tetrad experiments

Experiment 1 tested male assortative visual preference using two dummies each harboring the wings of a female of a different *M. helenor* subspecies. *M. h. theodorus* males did not show any preference for any female dummy, as shown by the probability of approaches and touches that were not different from the random 0.5 expectation (Binomial GLMM approaches: Estimate=-0.121±0.091 SE, $z=-1.332$, $p\text{-value}=0.183$, **Fig 4.A** [↗](#); touches: Estimate=1.600±1.094 SE, $z=1.462$, $p\text{-value}=0.144$, **Fig 4.B** [↗](#)). *M. h. bristowi* males, in contrast, approached and touched a conspecific *M. h. bristowi* dummy female significantly more often than expected by chance alone (Binomial GLMM approaches: Estimate=0.430±0.105 SE, $z=4.11$, $p\text{-value} < 0.0001$, **Fig 4.A** [↗](#); touches: Estimate=1.646±0.273 SE, $z=6.029$, $p\text{-value} < 0.0001$, **Fig 4.B** [↗](#)), suggesting a preference toward the *M. h. bristowi* females based on the recognition of visual cues alone.

Experiment 2 tested male pattern-based discrimination, using two dummies harboring the wings of *M. h. bristowi* females (i.e. same iridescence), but one of which was modified so as to present the thin blue bands found in *M. h. theodorus* (i.e. a different pattern). *M. h. bristowi* males tended to approach and touch significantly more often the wild-type *M. h. bristowi* female model as compared to the modified *M. h. bristowi* female model exhibiting a *theodorus* pattern (Binomial GLMMs approaches; Estimate=0.599±0.089 SE, $z=6.751$, $p\text{-value} < 0.0001$, touches; Estimate=2.805±0.944 SE, $z=2.972$, $p\text{-value} = 0.003$, see S8 Fig). As both female dummies shared the same blue iridescence, the choice of *M. h. bristowi* males was likely influenced by the difference in the pattern displayed by the dummy females, suggesting that the color pattern is used by males during mate choice.

Experiment 3 tested male color-based mate discrimination using two dummies harboring the wings of females from the two subspecies (i.e. different iridescence), but those of *M. h. bristowi* were modified so as to resemble *M. h. theodorus* (i.e. same pattern). *M. h. theodorus* males tended to approach and touch significantly more often the *M. h. theodorus* female wild-type model compared to the artificially modified, theodorus-like *M. h. bristowi* female dummy (Binomial GLMMs on approaches; Estimate=0.348±0.121 SE, $z=2.867$, $p\text{-value} = 0.004$, on touches; Estimate=1.172±0.465 SE, $z=2.521$, $p\text{-value} = 0.012$, see S9 Fig). As both dummy females shared the same color pattern, the preference expressed by *M. h. theodorus* males likely stemmed from the differences in the iridescent blue color, suggesting color-based discrimination in males.

In experiment 2 and 3, modified dummies were each time strongly rejected by males. Although all dummies contained traces of black marker to control for manipulation and treatment, we cannot entirely rule out that the artificial modifications performed on dummies could generate a specific visual rejection by males, because we cannot be sure whether butterflies perceive these modifications as equivalent to natural coloration. These experiments nevertheless suggest that visual cues – color pattern and iridescence – jointly contribute to mate choice in *M. helenor*.

Visual cues are not pre-zygotic reproductive barriers between the mimetic species *M. h. helenor* and *M. a. achilles*

We then tested whether these visual cues allowed males to express preferences toward their own species in the sympatric range of *Morpho achilles* and *M. helenor*. In a final experiment (experiment 4), we thus investigated whether males from the two species *M. achilles* and *M. helenor*, sampled in their sympatric range (French Guiana), have a visual preference toward their conspecific females: two female dummies harboring the wings of females from the two species were presented to each male. While none of the males (neither from *M. helenor* nor from *M. achilles*) touched the female models, they did approach them in the cages. The approach rate and the number of approaches did not significantly differ between *M. helenor* (rate = 61%; mean = 4.21, $sd = 5.34$) and *M. achilles* males (rate = 66%; mean = 7.18, $sd = 8.29$; binomial GLMM: $p\text{-value} = 0.626$), revealing similar levels of male courtship motivation. The independent analysis of the preference per species (**Fig 4.C** [↗](#)) shows that *M. helenor* males did not significantly approach their conspecific female more often than the heterospecific one (Binomial GLMM,

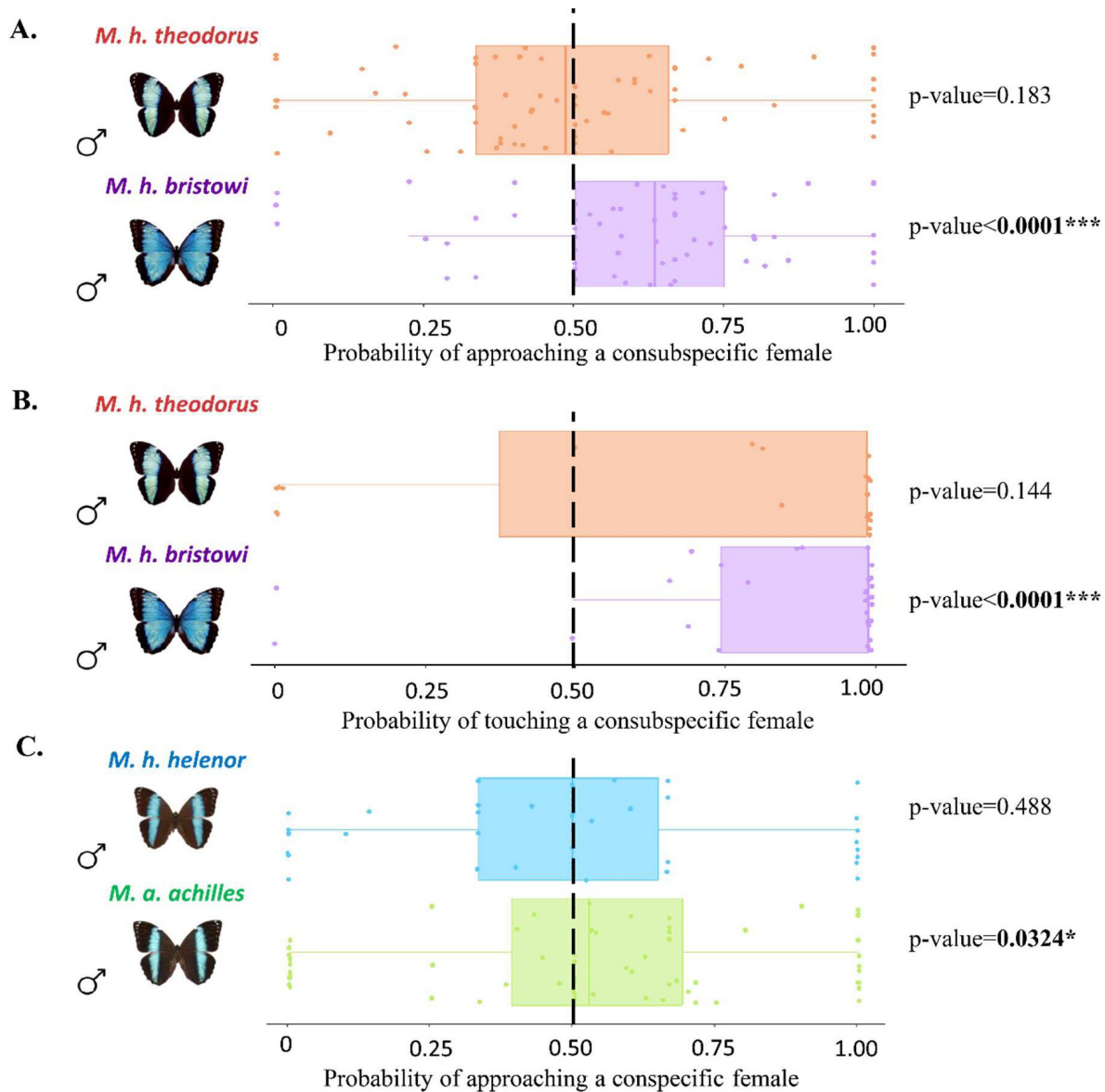


Fig 4

Morpho male preference based on visual cues alone.

Probabilities of (A) approaching and (B) touching a con-subspecific female for *M. h. bristowi* males (purple) and *M. h. theodorus* males (orange) from Ecuador, as well as the probabilities of (C) approaching a con-specific female for *M. h. helenor* males (blue) and *M. a. achilles* males (green) from French Guiana, measured during *Experiment 1*. The dotted line indicates the expected probability of approaching/touching a model if no preference is present. The $p\text{-values}$ resulting from the binomial GLMM testing for preferentially directed interactions toward one female dummy are shown next to the corresponding graphs.

Estimate=-0.161±0.198 SE, $z=-0.813$, $p\text{-value} = 0.416$), while *M. achilles* males did (Binomial GLMM, Estimate=0.196±0.092 SE, $z=2.14$, $p\text{-value} = 0.032$). However, note that the probability for *M. achilles* to approach one of its conspecifics is only 0.54, which is very low compared to the probability of *M. h. bristowi* to approach (0.61) or touch (0.84) its consubspecific in experiment 1. Visual discrimination thus appears to be lower among species, between the sympatric *M. helenor* and *M. achilles*, than within species, between the allopatric *M. helenor* subspecies, probably due to the remarkable similarity in color pattern and iridescence between the two species in their sympatric range.

Divergent chemical profiles observed between males of the sympatric species *M. helenor* and *M. achilles*

We finally investigated the volatile chemical compounds extracted from the genitalia of males and females from the mimetic sympatric species *M. helenor* and *M. achilles*. The nMDS analyses performed (Fig 5) suggested a strong divergence in the chemical profiles between males of the two species, especially for C16 to C30 compounds. This result contrasts with the strong similarity of chemical profiles found between the females of these two species in these longer-chain compounds (S10 Fig).

The PERMANOVA of the chemical spectra of *M. h. helenor* and *M. a. achilles* genitalia revealed a significant effect of sex, species ($p\text{-value} < 0.001$ on both the C8 to C16 and the C16 to C30 datasets, see Fig 5) and a significant interaction between the two variables. When analyzing the results for each sex separately (post-hoc), we found a significant effect of the species variable (PERMANOVA, $df=1$, $F=10.013$, $p\text{-value}=0.001$ for males and $df=1$, $F=1.928$, $p\text{-value}=0.049$ for females in the C8 to C16 compounds, PERMANOVA, $df=1$, $F=26.031$, $p\text{-value}=0.001$ for males and $df=1$, $F=2.776$, $p\text{-value} = 0.048$ for females in the C16 to C30 compounds), suggesting a substantial divergence between species in chemical bouquets, especially marked in males. Next, we performed an Indicator Value Analysis to identify the different compounds significantly associated with each species, separating males and females (Table S2). Among all the annotated compounds, we found traces of beta-ocimene, previously identified as a butterfly pheromone. Interestingly, different proportions of this compound were found on the genitalia of *M. helenor* and *M. achilles* males (Table S2).

Discussion

Convergent iridescence between sympatric *Morpho* butterflies suggests a prominent role of predation on the evolution of iridescence

Considering divergence time, wing iridescence is expected to be more different between the species *M. achilles* and *M. helenor* than among populations of *M. helenor*. Our results on the variation of iridescent properties of the blue patches of *M. helenor* and *M. achilles* show the opposite trend, with more divergence between populations within *M. helenor* than between *M. helenor* and *M. achilles* in their overlapping range. Together with the convergent evolution of the blue band width previously detected in multiple sympatric locations in these two species (Laurens et al., 2021), this observation is consistent with a primary role of natural selection on the evolution of wing coloration. Sympatric species are indeed submitted to the same abiotic conditions, favoring the evolution of similar adaptations (Chazot et al., 2014). For instance, wing coloration is involved in thermoregulation in butterflies, and similar environmental conditions may constrain the evolution of this trait. Despite this, thermal absorbance in iridescent species appears independent of environmental temperature, as experimentally measured in a large sample of Lepidopteran species (Bosi et al., 2008); similarly, recent experiments have shown

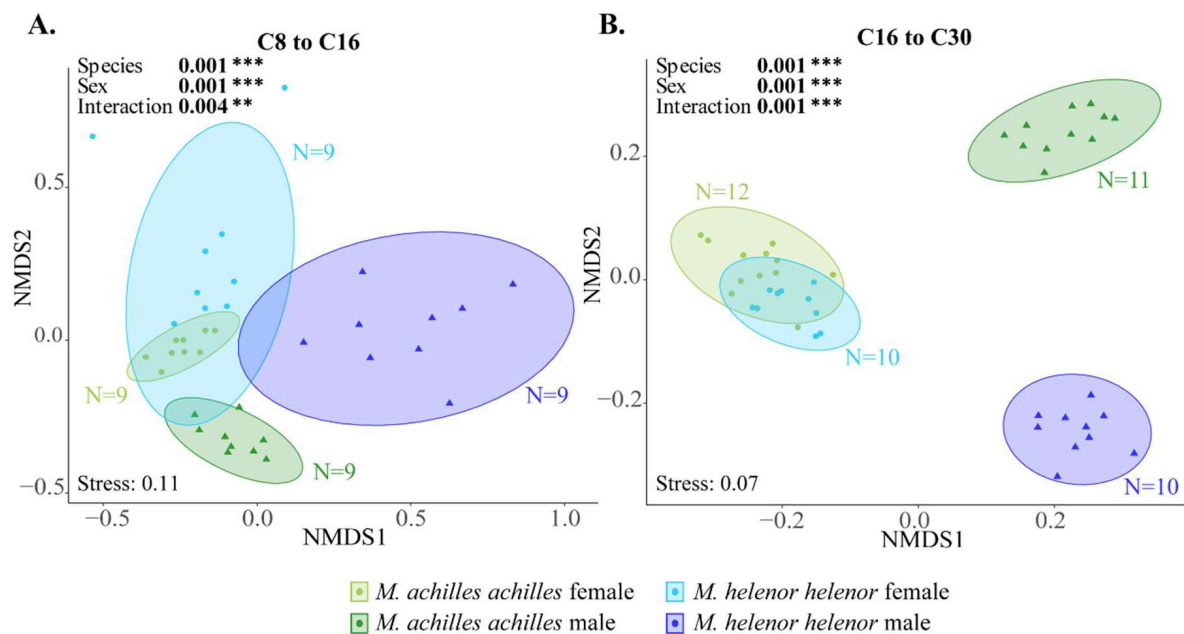


Fig 5

Divergent chemical compounds found on the genitalia of sympatric *Morpho* males.

nMDS representation of the differences in the concentration of (A) C8 to C16 chemical compounds and (B) C16 to C30 chemical compounds found in the genitalia of *M. h. helenor* and *M. a. achilles* males and females, calculated using Bray-Curtis distances. *M. h. helenor* are shown in blue (males are in dark blue and females in light blue), and *M. a. achilles* are shown in green (males are dark green and females are light green). The result of the PERMANOVA (999 permutations) testing the effect of sex and species on the chemical composition of *M. h. helenor* and *M. a. achilles* are shown on each figure.

that iridescent *Morpho* species are not better at thermoregulating than non-iridescent ones (Bouinier et al., 2025 [↗](#)), suggesting that the evolution of structural colors is not driven by selection on thermoregulation. Instead, the results of the visual models showing that these convergent iridescent signals are likely undistinguishable by birds supports an effect of selection by predators (Chouteau et al., 2016 [↗](#); Stuckert et al., 2014 [↗](#)). The lack of directionality for the iridescent signal of the wings further supports a limited effect of sexual selection driving divergence in visual signal. Directionality in iridescent signals is indeed expected when sexually selected traits incur a predation cost due to their conspicuousness: directionality might enhance signaling to mates while remaining poorly detected by predators (Endler, 1992 [↗](#)). Our study shows that *M. helenor* and *M. achilles* wings display a gradual variation of hue and brightness when changing the illumination/observation angles contrary to other iridescent butterfly species with more directional changes (e.g. *Hypolimnas bolina*; White et al., 2015 [↗](#)). Recent predation experiments showed that the gradual change of hue and brightness of iridescence can be involved in crypsis and camouflage: for instance, iridescent beetles are less detected by predators when placed against a leafy background (Kjernsmo et al., 2020 [↗](#); Thomas et al., 2023 [↗](#)). However, like most Satyrinae, *Morpho* butterflies tend to rest on leaves with their wings closed, thus hiding the iridescent patterns on the dorsal side of their wings at rest. Furthermore, iridescent species have a very erratic flapping flight, creating very conspicuous flashes (Le Roy, Amadori, et al., 2021; Young, 1971 [↗](#)). Gradual variations of hue and brightness during wing movements are thus consistent with the emission of a confusing visual signal that increases the chances of escaping predators. The convergence of visual signals could have been favored because of the indirect advantage that results in a reduction in attacks from bird predators that have experienced and learned to avoid the escape capabilities of these prey (Pinheiro & Freitas, 2014 [↗](#); Ruxton et al., 2004 [↗](#)). Coupled with results from predation experiments supporting an effect of dynamic flash coloration as a protective mechanism in *Morpho helenor* (Vieira-Silva et al., 2024 [↗](#)), convergence of color patterns in those sympatric butterflies is consistent with the hypothesis of escape mimicry, although this warrants further research.

Interestingly, our reflectance measurements showed that males are generally brighter than females in both species, implying that the flashes generated by males during flight could be more conspicuous. Field experiments have highlighted the territorial behavior of males found patrolling along rivers and displaying competitive interactions with other males (Le Roy, Roux, et al., 2021). This flight behavior strongly differs from female behavior, which are much less frequently observed in the wild and typically spend more time hidden in the understory (Young, 1973 [↗](#)). Considering that these ecological niche differences between males and females could result in different predation pressures in contrasted light conditions (open river banks vs. cluttered understory), iridescent signals could have evolved differently in the two sexes, and similarly so in both species (Allen et al., 2011 [↗](#); Reimchen & Nosil, 2004 [↗](#)).

Although we only compared one sympatric population and one allopatric population, our results on geographic variations in iridescence within *M. helenor* and similarity of iridescent coloration between *M. helenor* and *M. achilles* in sympatry are consistent with an effect of local selection exerted by predators. Further investigation on multiple populations is needed to test whether this trend can be generalized to multiple sympatric locations where other wing patterns can be observed. Additionally, characterizing convergence of iridescence is an indirect method to test for the effect of iridescence in predator deterrence: testing the predation rate of *Morpho* when presented with local or exotic predators would better characterize the direct effect of convergent iridescent in escaping predators in sympatry.

Iridescent color patterns can be used as mate recognition cues in *M. helenor*

Our tetrad and male mate choice experiments in *M. helenor* butterflies clearly show assortative mating based on wing coloration, mainly driven by the behavior of individuals from the subspecies *M. h. bristowi*. This result is consistent with the high attraction generated by blue coloration on males, as well as the preference of males for local vs. exotic wing color patterns found in previous field behavioral experiments carried out in Amazonian Peru (Le Roy, Roux, et al., 2021). Our male mate choice experiments suggest that both the wing patterning and the iridescent properties of the blue patch could have an effect on male preference. Furthermore, visual modeling also suggests that the differences of iridescent coloration between the populations of the same species can be perceived by mates. Coloration and color perception are key in mate recognition in many taxa (marine invertebrates; Baldwin & Johnsen, 2009 [↗](#), birds; Caro et al., 2021 [↗](#), mammals; Waitt et al., 2003 [↗](#)) but patterns can also inform mate choice (Houde, 1987 [↗](#); Pérez-Rodríguez et al., 2017 [↗](#)). In jumping spiders, both pattern and coloration are important cues used during mate choice (Zhou et al., 2021 [↗](#)). Similarly, in *Heliconius* butterflies, both pigmentary coloration and wing pattern are used in mate choice, although the preference is hierarchical and coloration is the most important cue (Finkbeiner et al., 2014 [↗](#)). Further experiments are needed to disentangle the relative effects of iridescent coloration and wing pattern in mate preference in *Morpho* butterflies, but our results show that iridescent color patterns can be used as mate recognition cues in *M. helenor*.

Evolution of visual and olfactory cues in similar sister-species living in sympatry

Interestingly, intraspecific variations in courtship motivation were detected in our behavioral experiments within the species *M. helenor*. Individuals from the western areas of the Andes displayed high courtship motivation while butterflies from the Amazonian area, where *M. helenor* coexists with *M. achilles*, showed a more limited response to our visual dummies. The joint convergence of both color pattern and blue iridescence between *M. helenor* and *M. achilles* in Amazonia probably makes visual cues poor species recognition signals: visual modeling even suggests that *M. helenor* cannot perceive the color difference between the two, whereas it could make the difference between two *M. helenor* subspecies displaying more divergent coloration. This is further supported by our male choice experiment showing that although *M. helenor* males living in sympatry with *M. achilles* are attracted by blue iridescent colors and approach each female, they do not preferentially interact with their conspecific female dummies. Nevertheless, *M. achilles* males tended to approach significantly more their conspecific females. Previous mate choice experiments performed in the wild suggested that *M. achilles* indeed preferred to fly around *M. achilles* females compared to *M. helenor* females (Le Roy, Roux, et al., 2021), suggesting that visual cues are still relevant in mate choice for some *Morpho* species living in sympatry with other *Morpho* butterflies. However, in our experiments, *M. achilles* males approached their conspecifics with a probability of 0.54 only, which is very low compared to what has been observed between intraspecific *M. helenor* subspecies, probably because of the high resemblance between *M. helenor* and *M. achilles* in sympatry.

The convergent evolution of iridescence in sympatric species might thus promote divergent evolution of alternative mating cues. By quantifying chemical compounds found on the genitalia of males and females sampled in Amazonian French Guiana, we did indeed find a strong divergence in the chemical profiles of sympatric *M. helenor* and *M. achilles*, especially in males. High male chemical divergence in sympatric species suggests the evolution of female mate choice based on olfactory discrimination (Mérot et al., 2015 [↗](#)). Although all compounds were not identified, some of those that were have previously been identified as pheromone compounds used during male/female interactions in butterflies. In particular, beta-ocimene was found on the

genitalia of males and is known to contribute to mate choice in courtship (Li et al., 2017 [↗](#)) or as an anti-aphrodisiac used by *Heliconius* males to repel other males after mating with a female (Schulz et al., 2008 [↗](#)). Additionally, we found strong divergence between males for heavier chemical compounds (C16-C30). Because cuticular hydrocarbons can be involved in mate discrimination during courtship (Ômura et al., 2020 [↗](#)), diverging non-volatile compounds could also be used as a discriminating cue. Although we did not perform mate choices to test for the discrimination of olfactory cue by females, divergence of male odorant cues could be due to reproductive interference between sympatric species (Bacquet et al., 2015 [↗](#); Dyer et al., 2014 [↗](#)). Altogether, our results therefore suggest that the convergent evolution of iridescent wing pattern between the sympatric *M. helenor* and *M. achilles* in Amazonia, likely promoted by shared predation pressures, may have negatively impact visual discrimination and favored the evolution of divergent olfactory cues. However, determining whether divergence of the chemical profiles of allopatric and visually more divergent *Morpho* butterflies occurs and to what extent would be needed in order to determine whether the divergence observed in sympatric *Morpho* species is truly due to reinforcement.

Conclusions

We found evidence of convergent iridescent patterns in sympatry suggesting that predation could play a major role in the evolution of iridescence. Further work is nevertheless needed to directly test this hypothesis and establish the importance of evasive mimicry in *Morpho* (Pinheiro et al., 2016 [↗](#)), similar to the Müllerian mimicry observed in chemically-defended species. There are however striking differences between the two types of mimicry: while a wide diversity of conspicuous color patterns are involved in Müllerian mimicry (Mallet & Gilbert, 1995 [↗](#)), the variation of iridescent patterns involved in evasive mimicry might be more tightly constrained, because iridescence itself directly participates in survival via visual confusion (Murali & Kodandaramaiah, 2020 [↗](#)). Moreover, a switch in color pattern is generally considered as a magic trait facilitating speciation in chemically defended species (Merrill et al., 2012 [↗](#)), and Müllerian mimicry frequently occurs among distantly-related species (Puissant et al., 2023 [↗](#)). Here we observe the co-existence of *Morpho* sister-species displaying convergent color patterns, suggesting that speciation should be promoted by the evolution of traits other than visual, such as olfactory chemical cues, or by the specialization into divergent ecological temporal niches (Le Roy, Roux, et al., 2021). Altogether, this study addresses how convergence in one trait as a result of biotic interactions may alter selection on traits in other sensory modalities, resulting in a complex mosaic of biodiversity.

Materials and Methods

I. Study system

We investigated trait variations in the sister-species *M. achilles* and *M. helenor*, that occur in sympatry in the Amazonian basin (Fig 1 [↗](#)), with two main sampling schemes, focusing on intraspecific and interspecific variations respectively.

A. M. helenor populations of Ecuador: Pupae from the Ecuadorian populations of *M. h. bristowi* and *M. h. theodorus* were purchased from a breeding farm located in Ecuador (Quinta De Goulaine, <https://quintadegoulaine.com/es/papillons.php> [↗](#)), frequently supplying their breeding with wild-caught individuals from Tena (Eastern Ecuador) and Pedro Vicente Madonado (Western Ecuador) respectively. No artificial selection was performed. These commercially-bought pupae were then raised in insectaries at STRI in Gamboa, Panama, between January and March 2023, and used for experiments 4 or 5 days after emergence to wait for sexual maturity.

Sympatric species of French Guiana: M. h. helenor and *M. a. achilles* individuals were wild-caught and raised in insectaries between July and September 2023 in the Amazonian forest of the Kaw Mountains (GPS coordinates: 4.57, -52.21), French Guiana, France.

II. Quantification and characterization of coloration and iridescence of *Morpho* butterfly wings

A. Measures of wing reflectance

We investigated the variations in iridescence in *M. h. bristowi*, *M. h. theodorus* (from Ecuador), *M. h. helenor* and *M. a. achilles* (from French Guiana), using 10 females and 10 males of each. We estimated the iridescence of the blue patches of the dorsal side of the wings by performing a series of reflectance measurements on wings positioned on a flat surface, following a template to ensure that the wing orientation remained the same across all measurements. Reflectance spectra captured at different angles were recorded on the right forewing, at a specific point located in the wing zone defined by the M3 and Cu1 veins according to the nymphalid ground plan (see [Martin & Reed, 2010](#) [DOI](#)). Wing reflectance was measured using a spectrophotometer (AvaSpec-ULS2048CL-EVO-RS 200-1100nm, Avantes) coupled with a deuterium halogen light source (AvaLight-DH-S-BAL, Avantes) for 21 combinations of illumination and observation angles, covering the proximo-distal and anteroposterior planes of the wing ([Fig 2.A](#) [DOI](#), see Supporting Information Text 1 for extended methods).

These measurements were repeated 3 times for all individuals to estimate the repeatability of the reflectance measurements at each angle using the R package *rptR* 0.9.22 ([Stoffel et al., 2017](#) [DOI](#)). Since no difference was detected between replicates for any of the tested angles (S12 Fig), the mean of these 3 measurements at each angle for each individual was used in the subsequent analyses.

B. Statistical analysis of iridescence

Each illumination/observation position produced a reflectance spectrum whose characteristics (*i.e.* distribution of reflectance) define the optical properties of the illuminated surface. We focused our analyses on the wavelength range [300, 700 nm], relevant for both butterfly ([Briscoe, 2008](#) [DOI](#)) and bird observers ([Bennett & Théry, 2007](#) [DOI](#)). All reflectance spectra were analyzed using the R package *pavo2* 2.9.0 ([Maia et al., 2019](#) [DOI](#)). The statistical analysis of both colorimetric variables and global iridescence were performed using the iridescence data of 10 males and 10 females from each population (*M. h. bristowi*, *M. h. theodorus*, *M. h. helenor* and *M. a. achilles*).

a. Extraction and analysis of colorimetric variables

To identify the main optical effect produced by the wing at each angle, we extracted three commonly-used colorimetric variables from each spectrum, using the *summary()* function in *pavo2*: namely Hue (color in the common meaning of the term - measured as the wavelength at the maximum reflectance), Brightness (average proportion of reflected light - measured by the mean reflectance over each spectra) and Chroma (also referred to as saturation, it characterizes the color's purity - measured as the ratio of the reflectance range of each spectra by the brightness). For each individual, we thus extracted these 3 variables for each of the 21 spectra obtained for different combinations of illumination/reflectance angles. Because the normality and heteroscedasticity assumptions were not met for the distributions of values in these three parameters, we relied on permutation-based ANOVAs (*vegan* 2.6.4 package, [Oksanen et al., 2001](#) [DOI](#)) to test the effect of sex, angle of measurement, locality and species on these three colorimetric variables. To specifically test whether differences in iridescence patterns were observed among species or localities, we estimated the interaction between the effects of *Morpho* populations and of the angle of illumination.

b. Global iridescence analysis

As iridescence broadly refers to any change in reflectance in relation to illumination and observation angle (Doucet & Meadows, 2009 [DOI](#)), we characterized the iridescence of each individual using all complete spectra obtained at all angular positions (400 wavelengths x 21 angular positions). We applied a Principal Component Analysis (PCA) to this dataset, where each dot corresponds to the iridescence of an individual wing and thereby showing the similarities and differences in iridescence among groups. The samples from Ecuador and French Guiana were analyzed separately to test for more precise differences in iridescence between allopatric and sympatric *Morpho* butterfly populations. Additionally, those comparisons correspond to the phenotypes used in the mate choice experiments (see below). The coordinates of the 10 first PCs (representing 98% of the explained variance) were then extracted and a PERMANOVA was performed to test for the effect of sex, locality and species on iridescence patterns.

III. Modeling how *Morpho* butterflies and avian predators perceive *M. helenor*

To investigate whether the variations observed among the different spectra could be perceived by different observers, we then used the R package *pavo2* 2.9.0 (Maia et al., 2019 [DOI](#)) to model both avian and butterfly visual capabilities. We used the avian UV-sensitive vision model implemented in *pavo* (“*avg.uv*”) as a proxy for predator vision and specifically built a model of *M. helenor* vision, using the spectral sensitivity measured in Pirih et al., 2022 [DOI](#). We built a hexachromatic visual model using the *sensmodel* function, using 345, 446, 495, 508, 570 and 602 nm as the maximum of absorbance in our six simulated cones respectively (S11 Fig). The *vismodel* function was used to compute the quantum catch of each receptor in the *Morpho* vision model under standard daylight illumination for each spectrum. Finally, assuming that the discrimination of colors is limited by photoreceptor noise (Receptor-Noise Limited model, Vorobyev et al., 2001 [DOI](#)), we estimated the distance between colored stimuli as perceived by Morphos using the *bootcoldist* function. We used the default options (*weber* = 0.1, *weber.achro* = 0.1) and set the photoreceptor densities for each cones used in the visual model to 1. The resulting chromatic distances (*dS*), achromatic distances (*dL*) and confidence intervals were then used to test how much the visual capacities of *Morpho* butterflies could allow the discrimination of wing reflectance (brightness for achromatic distances, hue for chromatic distances) depending on sex, localities and species, at each angle. Similarly, we used the avian UV sensitive visual model implemented in *pavo* to test the discrimination of *Morpho* wing reflectance by predators. We used *dS* ≥ 1 *Just Noticeable Difference* as the threshold of discrimination.

IV. Mating experiments

All mate choice experiments took place in cages of similar dimensions (3m x 3m x 2m), in which *Morpho* butterflies, especially males, were able to fly freely. All *Morpho* butterflies were fed 30 minutes before the beginning of the experiments taking place between 8am and 12am.

a. Testing for mate discrimination within *M. helenor*

To study the discrimination of mating partners within the species *M. helenor*, tetrad experiments were performed between *M. h. bristowi* and *M. h. theodorus*. For each experiment, one virgin female and one mature male of both *M. h. bristowi* and *M. h. theodorus* were placed within the same cage. Their starting positions in each corner of the cage were randomized. The first mating occurring out of the four possibilities was recorded and the experiment was stopped as soon as the male joined their genitalia. Experiments could also be ended when no mating occurred within 5 hours of trial. We carried out N = 33 trials to estimate whether the first mating observed

preferentially involved assortative pairs, due to either male and/or female mate preference, of which N=30 trials actually resulted in assortative or disassortative matings. We statistically tested for departure from the random mating expectation using a Fisher's exact test.

Note that *M. achilles* is not commercially bred and wild females are usually all mated, preventing experiments on virgin females in this species, and thus restricting experiments on virgin females in the commercially-bred species *M. helenor*.

b. Identifying specific visual cues used during mate choice by *M. helenor* males

As male *Morpho* tend to patrol in the wild to find females, and because reports show that males are attracted to bright blue coloration in the wild (Le Roy, Roux, et al., 2021), we specifically tested for male mate choice using visual cues. Females are also likely to be sensitive to color pattern in mate recognition. However, we did not test for female choice because their behavior was not compatible with experiments involving flying around butterfly dummies: female individuals tended to hide away from the males in our experimental cages. To investigate the visual preference of *M. helenor* males for the dorsal color pattern of females, we performed several choice experiments using dummy butterflies built with actual female wings. To ensure that preferences were not triggered by olfactory cues, all female wings used to build the artificial dummies were washed with hexane before any experiment. The details of the experimental design and specific visual cues tested are described in **Figure 6** [↗](#).

First, we tested male preference for the visual appearance of *M. h. bristowi* and *M. h. theodorus* females (Experiment 1). Since *M. h. bristowi* and *M. h. theodorus* females differ both in terms of color patterns (width of blue band) and iridescence, we then conducted experiments to isolate these two components.

We specifically tested the effect of the proportion of blue to black areas (Experiment 2) by presenting males *M. h. bristowi* to female dummies harboring *M. h. bristowi* wings, except one was modified with black ink to resemble the narrow blue band on *M. h. theodorus* female wings.

We finally tested the effect of the blue iridescent band within the same black and blue pattern (Experiment 3) by presenting *M. h. theodorus* males to a *M. h. theodorus* female with a narrow blue band, and to the modified *M. h. bristowi* female dummy with a narrow band. The wing pattern modifications were performed using a black permanent marker.

To ensure that ink odor would not bias the behavior of *Morpho* males, all dummies used in experiments 2 and 3 contained traces of black marker (either on the modified bands or on the ventral side of the wings of natural females). Nevertheless, we cannot entirely discard the specific visual effect of the marker on the dorsal face.

In all experiments, two female dummies were hanging from the ceiling of the cage, 1.5 m apart, and presented to a live male. The male was released in the experimental cage in front of the two female dummies and was free to interact with them for 10 minutes. The positions of the female models were swapped 5 minutes into each trial to avoid potential position-related bias. Two types of male behaviors were recorded. Each time a male flew within a 40 cm radius from a female model, an "approach" was recorded. Each time a male touched a female dummy, a "touch" was recorded. The number of "approaches" and "touches" toward each female dummy per trial was used to determine male preference. Each male performed this experiment 3 times on 3 different days to account for potential individual variations in mate preference.

Experiments (Location)	Dummy female 1	Dummy female 2	Live males	Test
1 (STRI)	<i>M. h. theodorus</i> 	<i>M. h. bristowi</i> 	<i>M. h. theodorus</i> (n = 33)	Assortative preference within species
			<i>M. h. bristowi</i> (n = 28)	
2 (STRI)	<i>M. h. bristowi</i> 	<i>M. h. bristowi</i> with thinner blue band 	<i>M. h. bristowi</i> (n = 28)	Discrimination on the pattern
3 (STRI)	<i>M. h. theodorus</i> 	<i>M. h. bristowi</i> with thinner blue band 	<i>M. h. theodorus</i> (n = 33)	Discrimination on the iridescent blue
4 (French Guiana)	<i>M. h. helenor</i> 	<i>M. a. achilles</i> 	<i>M. h. helenor</i> (n = 28)	Assortative preference between species
			<i>M. a. achilles</i> (n=37)	

Figure 6

Experimental design of the male choice experiments

performed between the Morphos originating from Ecuadorian populations of *M. helenor* (eastern population: *M. h. theodorus* and western population: *M. h. bristowi*, experiments 1 to 3) and from French Guiana (*M. h. helenor* and *M. a. achilles*, experiment 4), along with the associated preferences and visual cues tested. Note that the same males were used for experiments 1, 2 and 3.

c. Testing whether visual cues can act as a pre-zygotic barrier between *M. h. helenor* and *M. a. achilles*

In order to investigate the importance of color vision in species recognition, we then performed a male choice experiment with *M. h. helenor* and *M. a. achilles* (experiment 4 in [Figure 6](#)) in French Guiana, using natural wings of females *M. h. helenor* and *M. a. achilles* as artificial butterflies following the exact same protocol above.

a. Statistical analyses

Comparing male mating motivation

Because we noticed behavioral differences between the species and subspecies of *Morpho* during the course of the experiments 1-4, we attempted to account for variations in male mating motivation. The approach and touch rates of males, *i.e.* their willingness to approach and touch (respectively) any female dummies during each trial, was defined as the proportion of trials where the males approached and touched (respectively) either female dummies at least once. The approach rate and touch rate of *M. h. bristowi*, *M. h. theodorus*, *M. h. helenor* and *M. a. achilles* males were calculated, and a binomial generalized linear model (GLMM) was used to test whether the approach or touch rate was significantly different between males from different populations.

All statistical analyses were performed with the R package *glmmTMB* (Brooks et al., 2017). The normality, homogeneity of variance and overdispersion tests for the GLMMs were performed using the R package *DHARMA* (Hartig, 2022). We controlled for potential individual variations by using the male ID and trial number as a random effect in all the tests.

Visually-based male mate preferences

We then tested for male preferential attraction for different dummy females. For each male individual, we recorded the number of approaches and touches directed toward the two female dummies and modeled this as a two-column response variable. Similar to the protocol above, we used a binomial GLMM to test whether individuals preferentially interacted with dummy 1 vs. dummy 2, and added a random effect for individual identity to account for repeated observations from the same individual. A significant signal was detected when the probability of touching a dummy was different from the 0.5 proportion expected under random mating, and interpreted as a signal of preference.

V. Pheromone analysis for sympatric species

a. Chemical analyses

We investigated variations in chemical compounds between the two sympatric species *M. helenor* and *M. achilles* sampled in French Guiana. The male and female genitalia of wild caught *M. h. helenor* and *M. a. achilles* individuals (from 9 up to 12 per sex per species) were dissected and extracted in 20 μ L of dichloromethane, containing 5 μ L of octadecane as an internal standard. Two sets of chromatography were performed in order to extract either the light compounds (all volatile carbonate compounds below C16) or the heavier ones (all carbonate compounds above C16) found on the genitalia of those Morphos.

The chemical compounds of those samples were analyzed by gas chromatography and time-of-flight mass spectrophotometry (GC-ToF-MS) using a 8890 GC system (Agilent, Santa Clara, USA) coupled to a PEGASUS BT mass spectrophotometer (LECO, Saint Joseph, USA). The connected capillary column was a Rxi-5ms (30 m x 0.25 mm, $df = 0.25 \mu$ m, Restek, Bellefonte USA) and helium was used as carrier gas at 1 mL min⁻¹ constant flow rate. 1 μ L of each sample was injected in a GC

split/splitless injector at 300 °C. Two temperature programs were used: (i) the analysis of the C8-C16 compounds started at 40 °C maintained for 1 min, then increased at 210°C at 6 °C min⁻¹ rate, finally increased to 340 °C at a 10 °C min⁻¹, and (ii) the analysis of the C16-C30 compounds started at 70 °C maintained for 1 min, then increased to 150 °C at 30 °C min⁻¹ rate and finally increased to 340 °C at a 6 °C min⁻¹ rate. Mass spectra were recorded with a 70eV electron impact energy. Mixtures of alkanes C8 to C19 and C16 to C44 (ASTM® D5442 C16-C44 Qualitative Retention Time Mix, Supelco, Bellefonte, USA) were injected under the same conditions to be used as external standards.

The chromatograms were converted in netCDF format and operated using *MZmine 2.43* for compound detection and area integration. To this aim, chromatograms were cropped before 4 min and after 27 min, and baseline-corrected using the Rolling ball algorithm (Kneen & Annegarn, 1996 [\[4\]](#)). Mass detection was performed using the centroid mass detector, and chromatograms of each mass were constructed using the ADAP algorithm (Myers et al., 2017 [\[4\]](#); Pluskal et al., 2010 [\[4\]](#)). Chromatograms were deconvoluted into individual peaks using the local minimum search algorithm and aligned on a master peak list with the RANSAC algorithm. Finally, the Peak Finder gap filling algorithm was used to search for missing peaks, and a matrix compiling the area under each peak (*i.e.* the concentration of each component) detected for each butterfly sample was extracted. This matrix was purged from siloxane like compounds as well as compounds present in less than 10% of the samples. The areas of the blanks were subtracted to the samples, and the final concentration data was used for statistical analysis. Mass spectra were then visualized using the ChromaTOF® software, compared to MZmine-selected ones, in order to manually annotate the chemical compounds. Annotation were based on scan comparison with the NIST2017 database, and linear retention indexes (LRI) calculated from the alkane mixtures with free database (<https://pubchem.ncbi.nlm.nih.gov/> [\[4\]](#)) and literature (Adams, 2017).

b. Statistical analyses of the chemical compounds

In order to quantify variations in chemical compounds detected in the different sample, we performed a nonmetric multidimensional scaling (NMDS) ordination using the Bray-Curtis dissimilarity calculation on the concentrations extracted from the chromatograms. By considering the amount of each compound found in our samples as a variable, we performed a permutation MANOVA with the *adonis2()* function found in the *vegan 2* package on R. We tested whether the sex or the species had a significant effect on the chemical bouquet produced by their genitalia.

We also performed an Indicator Value Analysis to identify the chemical compounds significantly associated to our different *Morpho* samples with the *mutlipatt* function from the *indicspecies* package in R using the IndVal index from (Dufrêne & Legendre, 1997 [\[4\]](#)).

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Additional files

Supporting information [↗](#)

Additional information

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Joint Public Review:

Summary:

Ledamoisel et al. examined the evolution of visual and chemical signals in closely related *Morpho* butterfly species to understand their role in species coexistence. Using an integrative, state-of-the-art approach combining spectrophotometry, visual modeling, and behavioral mate choice experiments, they quantified differences in wing iridescence and assessed its influence on mate preference in allopatry and sympatry. They also performed chemical analyses to determine whether sympatric species exhibit divergent chemical cues that may facilitate species recognition and mate discrimination. The authors found iridescent coloration to be similar in sympatric *Morpho* species. Furthermore, male mate choice

experiments revealed that in sympatry, males fail to discriminate conspecific females based on coloration, reinforcing the idea that visual signal convergence is primarily driven by predation pressure. In contrast, the divergence of chemical signals among sympatric species suggests their potential role in facilitating species recognition and mate discrimination. The authors conclude that interactions between ecological pressures and signal evolution may shape species coexistence.

Strengths:

The study is well-designed and integrates multiple methodological approaches to provide a thorough assessment of signal evolution in the studied species. We appreciate the authors' careful consideration of multiple selective pressures and their combined influence on signal divergence and convergence. Additionally, the inclusion of both visual and chemical signals adds an interesting and valuable dimension to the study, enhancing its importance. Beyond butterflies, this research broadens our understanding of multimodal communication and signal evolution in the context of species coexistence.

Reviewing Editor comment:

The authors have improved their submission after revisions and responded to the previous concerns of the reviewers.

<https://doi.org/10.7554/eLife.106098.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

In this study, Ledamoisel et al. examined the evolution of visual and chemical signals in closely related Morpho butterfly species to understand their role in species coexistence. Using an integrative, state-of-the-art approach combining spectrophotometry, visual modeling, and behavioral mate choice experiments, they quantified differences in wing iridescence and assessed its influence on mate preference in allopatry and sympatry. They also performed chemical analyses to determine whether sympatric species exhibit divergent chemical cues that may facilitate species recognition and mate discrimination. The authors found iridescent coloration to be similar in sympatric Morpho species. Furthermore, male mate choice experiments revealed that in sympatry, males fail to discriminate conspecific females based on coloration, reinforcing the idea that visual signal convergence is primarily driven by predation pressure. In contrast, the divergence of chemical signals among sympatric species suggests their potential role in facilitating species recognition and mate discrimination. The authors conclude that interactions between ecological pressures and signal evolution may shape species coexistence.

Strengths:

The study is well-designed and integrates multiple methodological approaches to provide a thorough assessment of signal evolution in the studied species. I appreciate the authors' careful consideration of multiple selective pressures and their combined influence on signal divergence and convergence. Additionally, the inclusion of both visual and chemical signals adds an interesting and valuable dimension to the study, enhancing its importance. Beyond butterflies, this research broadens our understanding of multimodal communication and signal evolution in the context of species coexistence.

Weaknesses:

(1) The broader significance of the findings needs to be better articulated. While the authors emphasize that comparing adaptive traits in sympatry and allopatry provides insights into selective processes shaping reproductive isolation and coexistence, it is unclear what key conceptual or theoretical questions are being addressed. Are these patterns expected under certain evolutionary scenarios? Have they been empirically demonstrated in other systems? The authors should explicitly state the overarching research question, incorporate some predictions, and better contextualize their findings within the existing literature. If the results challenge or support previous work, that should be highlighted to strengthen the study's importance in a broader context.

We thank the reviewer for their valuable feedback. We understand that the framing of the results and the discussion may fail to convey the broader significance of our findings. In the first version of the manuscript, we framed our manuscript around the processes shaping reproductive isolation and co-existence in sympatry, but now realize that this question was too broad in regards to our results. We thus strictly focused on outlining the importance of ecological interactions in the evolution of traits in sympatric species. In the revised version of the manuscript, we rewrote the first paragraph of the introduction to introduce context regarding the effect of ecological interactions on trait evolution (lines 43-60). We then explicitly introduce the theoretical question investigated in our paper (i.e. “we investigate how ecological interactions in sympatry can constrain natural and sexual selection shaping trait evolution”, lines 62-63) and our predictions regarding the evolution of traits in sympatry vs. allopatry (lines 74-80). We also added predictions regarding our experiments on *Morpho* at the end of the introduction (lines 146-157). As a result, the discussion is now better aligned with the introduction, by discussing the putative effect of predation and mate choice on the evolution of wing iridescence in *Morpho*.

(2) The motivation for studying visual signals and mate choice in allopatric populations (i.e., at the intraspecific level) is not well articulated, leaving their role in the broader narrative unclear. In particular, the rationale behind experiments 1, 2, and 3 is not well defined, as the authors have not made a strong case for the need for these intraspecific comparisons in the introduction. This issue is further compounded by the authors' primary focus on signal evolution in sympatry throughout both the results and the discussion. For instance, the divergence of iridescence in allopatry is a potentially interesting result. But the authors have not discussed its implications.

We now clearly state in the introduction our motivation for studying visual signals and mate choice in allopatric populations (lines 74-80, lines 146-157). We argued that intraspecific comparisons help identify whether visual cues can be used in mate recognition between phylogenetically close subspecies, between whom visual resemblance is supposed to be higher than between closely-related species (tetrad experiment, and experiment 1). As *M. h. bristowi* and *M. h. theodorus* have different wing pattern, we also used this comparison to identify the traits involved in male mate preference within a species, testing the importance of iridescent color (experiment 2) or iridescent patterning (experiment 3). The results of those experiments can then be used to assess whether these traits are used in species recognition between sympatric species. See also our answers to recommendations 11 and 15 from reviewer #1.

Overall, given that the primary conclusions are based on results and analyses in sympatry, the role of allopatric populations in shaping these conclusions needs to be better integrated and justified. Without a stronger link between the comparative framework and the study's key takeaways, the use of allopatric populations feels somewhat peripheral rather than central to the study's aim. Since the primary

conclusions remain valid even without the allopatric comparisons, their inclusion requires a clearer rationale.

To make a stronger case for the use of the allopatric population in our manuscript, we strengthened the justification behind the study of intraspecific allopatric populations vs. interspecific sympatric populations, as the iridescence measurements and the mate choice experiments in allopatric populations can serve as a baseline in studying how species interactions can shape the evolution of traits and mate recognition when compared to sympatric populations. Following your major comment #1, we rewrote the introduction to include a justification to the need for studying allopatric vs. sympatric populations (lines 74-80), and also further highlighted the need to study iridescence in sympatric species to fully understand the trait evolution of sympatric species in the discussion (339-343).

(3) While the authors demonstrate that iridescence is indistinguishable to predators in sympatry, they overstate the role of predation in driving convergence. The present study does not experimentally demonstrate that iridescence in this species has a confusion effect or contributes to evasive mimicry. Alternatively, convergence could result from other selective forces, such as signal efficacy due to environmental conditions, rather than being solely driven by predation.

We acknowledge that our study does not directly demonstrate that iridescence contributes to evasive mimicry. We did tone down the interpretation of the results in the discussion and state that predation is not the only selective pressure that could have promoted a convergent evolution of iridescence in sympatric species, as iridescence is a trait that could be involved in thermoregulation (lines 346-353) and camouflage (lines 363-369) for example. We made sure to mention that convergence in iridescent signals in sympatry is only an indirect support to the evasive mimicry hypothesis, and that further research is still needed, including direct predation experiments, to show that this convergence is indeed triggered by predation (lines 391-396).

Reviewer #2 (Public review):

*This study presents an investigation of the visual and chemical properties and mating behaviour in *Morpho* butterflies, aimed at addressing the nature of divergence between closely related species in sympatry. The study species consists of three subspecies of *Morpho helenor* (*bristowi*, *theodorus*, and *helenor*), and the conspecific *Morpho achilles achilles*. The authors postulate that whereas the iridescent blue signals of all (sub)species should function as a predator reduction signal (similar to aposematism) and therefore exhibit convergence, the same signals should indicate divergence if used as a mating signal, particularly in sympatric populations. They also assess chemical profiles among the species to assess the potential utility of scent in mediating species/sex discrimination.*

*The authors first used reflectance spectrometry to calculate hue, brightness, and chroma, plus two measures of "iridescence" (perhaps better phrased as angular dependence) in each (sub)species. This indicated the ubiquitous presence of sexual dimorphism in brightness (males brighter), which also appears to be the case for iridescence (Figure 3A-B). Analysis of these data also indicated that whereas there is evidence for divergence among subspecies in allopatry, the same evidence is lacking for species in sympatry ($P = 0.084$). This was supported further by visual modelling, which showed that both conspecifics and birds should be (theoretically) capable of perceiving the colour difference among allopatric populations of *M. helenor*, whereas the same is not true for the sympatric species.*

The authors then conducted mate choice trials, first using live individuals and second using female dummies. The live experiments indicated the presence of assortative mating

*among the two subspecies of *M. helenor* (*bristowi* and *theodorus*). The dummy presentations indicated (a) *bristowi* males prefer conspecific wings, whereas *theodorus* have no preference, (b) *bristowi* males prefer the con(sub)specific colour pattern, (c) *theodorus* prefer the con(sub)specific iridescence when the pattern is manipulated to be similar among female dummies. A fourth experiment, using sympatric *M. achilles* and *M. helenor*, indicated no preference for conspecific female dummies. Finally, chemical analysis indicated substantial differences between these two species in putative pheromone compounds, and especially so in the males.*

*The authors conclude that the similarity of iridescence among species in sympatry is suggestive of convergence upon a common anti-predation signal. Despite some behavioural evidence in favour of colour (iridescence)-based mate discrimination, chemical differences between *Achilles* and *Helenor* are posed as more likely to function for species isolation than visual differences.*

Overall, I enjoyed reading this manuscript, which presents a valiant attempt at studying visual, chemical and behavioural divergence in this iconic group of butterflies.

Major comments

*My only major comment concerns the authors' favoured explanation for aposematism (or evasive mimicry) for convergence among species, which is based upon the you-can't-catch-me hypothesis first presented by Young 1971. Although there is supporting work showing that iridescent-like stimuli are more difficult to precisely localize by a range of viewers, most of the evidence as applied to the *Morpho* system is circumstantial, and I'm not certain that there is widespread acceptance of this hypothesis. Given that the present study deals with closely-related (sub)species, one alternative explanation - a "null" hypothesis of sorts - is for a lack of divergence (from a common starting point) as opposed to evolutionary convergence per se. In other words, two subspecies are likely to retain ancestral character states unless there is selection that causes them to diverge. I feel that the manuscript would benefit from a discussion of this alternative, if not others. Signalling to predators could very well be involved in constraining the extent of convergence, but this seems a little premature to state as an up-front conclusion of this work. There is also the result of a *dorsal* wing manipulation by Vieira-Silva et al. 2024 which seems difficult to reconcile in light of this explanation. Whereas this paper is cited by the authors, a more nuanced discussion of their experimental results would seem appropriate here.*

We thank the reviewer for their constructive comments on our manuscript. We appreciate the reviewer's concern regarding the way iridescence convergence between sympatric species is discussed in our manuscript, which align with similar concerns raised by Reviewer 1. Indeed, the you-can't-catch-me hypothesis has not been yet empirically tested in *Morpho*, this is currently a working hypothesis only supported by indirect lines of evidence.

Among the 30 known *Morpho* species, iridescence is most likely the ancestral character, notably because iridescence is a trait shared by a majority of *Morpho* (we now mention this in the introduction lines 108-110). In this paper, we thus did not aim to identify the evolutionary forces involved in the appearance of iridescence in this group, but rather wanted to understand to what extent ecological interactions can impact the diversification (or not) of this trait. As such, the dorsal manipulations performed in Vieira-Silva et al 2024 showing that iridescence in *Morpho* may have a similar effect than crypsis does not impact our working hypothesis. Instead, we use Vieira-Silva et al 2024 to discuss the potential anti-predator effect of iridescence, that could potentially promote convergent evolution of iridescent patterns.

In the main text, we now clearly mention our null hypothesis: under a scenario of neutral evolution of iridescence, we would expect that the divergence in wing coloration between two *M. helenor* subspecies would be lower than between two different *Morpho* species (*M. helenor* and *M. achilles*) and showed that our results sharply differ from this null expectation.

We then improved the discussion by adding alternative hypotheses potentially explaining the convergent iridescent signal detected in sympatric species: we discussed the expected effect under neutral evolution (lines 339-343), but also added alternative hypotheses regarding the diversification of iridescence due to camouflage (lines 363-369), predator evasion (lines 373-377) and thermoregulation (lines 346-353).

Reviewer #3 (Public review):

The authors investigated differences in iridescence wing colouration of allopatric (geographically separated) and sympatric (coexisting) Morpho butterfly (sub)species. Their aim was to assess if iridescence wing colouration of Morpho (sub)species converged or diverged depending on coexistence and if iridescence wing colouration was involved in mating behaviour and reproductive isolation. The authors hypothesize that iridescence wing colouration of different (sub)species should converge in sympatry and diverge in allopatry. In sympatry, iridescence wing colouration can act as an effective antipredator defence with shared benefits if multiple (sub)species share the same colouration. However, shared wing colouration can have potential costs in terms of reproductive interference since wing colouration is often involved in mate recognition. If the benefits of a shared antipredator defence outweigh the costs of reproductive interference, iridescence wing colouration will show convergence and alternative mate recognition strategies might evolve, such as chemical mate recognition. In allopatry, iridescence wing colouration is expected to diverge due to adaptation to different local conditions and no alternative mate recognition is expected.

Strengths:

- (1) Using allopatric and sympatric (sub)species that are closely related is a powerful way to test evolutionary hypotheses*
- (2) By clearly defining iridescence and measuring colour spectra from a variety of angles, applying different methods, a very comprehensive dataset of iridescence wing colouration is achieved.*
- (3) By experimentally manipulating wing coloration patterns, the authors show visual mate recognition for *M. h. bristowi* and could, in theory, separate different visual aspects of colouration (patterns VS iridescence strength).*
- (4) Measurements of chemical profiles to investigate alternative mate recognition strategies in case of convergence of visual signals.*

Weaknesses:

In my opinion, studies should be judged on the methods and data included, and not on additional measurements that could have been taken or additional treatments/species that should be included, since in most ecological and evolutionary studies, more measurements or treatments/species can always be included. However, studies do need to ensure appropriate replication and appropriate measurements to test their hypothesis AND support their conclusions. The current study failed to ensure appropriate replication, and in various cases, the results do not support the conclusions.

First, when using allopatric and sympatric (sub)species pairs to test evolutionary hypotheses, replication is important. Ideally, multiple allopatric and sympatric (sub)species pairs are compared to avoid outlier (sub)species or pairs that lead to biased conclusions. Unfortunately, the current study compares 1 allopatric and 1 sympatric (sub)species pair, hence having poor (no) replication on the level of allopatric and sympatric (sub)species pairs,

We would like to thank the reviewer for their constructive feedback. We agree that replication is important to test evolutionary hypotheses and that our study lacks replication for allopatric and sympatric *Morpho* populations. Ideally, one would require several allopatric and sympatric replicates to conclude on the effect of species interaction in trait evolution. Our study is a preliminary attempt at answering this question, covering a few *Morpho* populations but proposing a broad assessment of iridescence and mate preference for those populations. We clearly mentioned in the discussion that investigating multiple populations is needed to test whether the trend we observed in this paper can be generalized (line 388-392).

Second, chemical profiles were only measured for sympatric species and not for allopatric (sub)species, which limits the interpretation of this data. The allopatric (sub)species could have been measured as non-coexistence "control". If coexistence and convergence in wing colouration drives the evolution of alternative mate recognition signals, such alternative signals should not evolve/diverge for allopatric (sub)species where wing colouration is still a reliable mate recognition cue. More importantly, no details are provided on the quantification of butterfly chemical profiles, which is essential to understand such data. It is unclear how the chemical profiles were quantified and what data (concentrations, ratios, proportions) were used to perform NDMS and generate Figure 5 and the associated statistical tests.

We recognize that having the chemical profiles of the genitalia of the *Morpho* from the allopatric populations would have made a stronger case in favor of reinforcement acting on the divergence of the chemical compounds found on the genitalia of the sympatric *Morpho* species. Due to limited access to the biological material needed at the time of the chromatography, we could not test for lower divergence in the chemical profiles of allopatric *Morpho* butterflies. We made sure to mention this limitation in the discussion (lines 457-461).

We already stated in the methods that we compiled the area under the peak of each components found in the chromatograms of our samples and that we performed all the statistical analyses on this dataset. To make it clearer, we mention in the new version of the manuscript that the area under the peak of each component allows to measure the concentration of the components (in the methods lines 720, 723, 733). We also added some precisions in the legend of Figure 5.

Third, throughout the discussion, the authors mention that their results support natural selection by predators on iridescent wing colouration, without measuring natural selection by predators or any other measure related to predation. It is unclear by what predators any of the butterfly species are predated on at this point

We made sure to mention in the introduction (line 132-136) and in the discussion (line 373-377) that previous predation experiments performed on *Morpho* and other butterflies showed evidence that birds are likely predators for these species. These observations lead us to test for the putative effect of predation on the evolution of their color pattern, without directly testing predatory rates. We made sure this information is transparent in the revised manuscript, and now precise that assessing wing convergence is only an indirect way of testing the escape mimicry hypothesis (line 393-396).

To continue on the interpretation of the data related to selection on specific traits by specific selection agents: This study did not measure any form of selection or any selection agent. Hence, it is not known if iridescent wing colouration is actually under selection by predators and/or mates, if maybe other selection agents are involved or if these traits converge due to genetic correlations with other traits under selection. For example, Iridescent colouration in ground beetles has functions as antipredator defence but also thermo- and water regulation. None of these issues are recognized or discussed.

The lack of discussion of alternative selective pressures involved in the evolution of iridescence was pointed out by all reviewers. We thus modified the text to account for this comment, and no longer limit our discussion to the putative effects of predation. We now specifically discuss alternative hypotheses, including crypsis (362-369) and thermoregulation (line 346-353).

Finally, some of the results are weakly supported by statistics or questionable methodology.

Most notably, the perception of the iridescence coloration of allopatric subspecies by bird visual systems. Although for females, means and errors (not indicated what exactly, SD, SE or CI) are clearly above the 1 JND line, for males, means are only slightly above this line and errors or CIs clearly overlap with the 1 JND line. Since there is no additional statistical support, higher means but overlap of SD, SE or CI with the baseline provides weak statistical support for differences.

We thank the reviewer for bringing interpretation issues concerning the chromatic distances of allopatric *Morpho* species measured with a bird vision model. We made sure to be nuanced in the description of this graph in the results section (line 208-212). Note that this addition does not change our main conclusion stating that *Morpho* and predator visual models better discriminate iridescence differences between allopatric subspecies than between sympatric species.

We now also clearly mention in the figure's legend that the error bars represent the confidence intervals obtained after performing a bootstrap analysis, in addition to the mention of the nature of the error bars already mentioned in the methods (line 580).

*Regarding the assortative mating experiment, the results are clearly driven by *M. bristowi*. For *M. theodorus*, females mate equally often with conspecifics (6 times) as with *M. bristowi* (5 times). For males, the ratio is slightly better (6 vs 3), but with such low numbers, I doubt this is statistically testable. Overall low mating for *M. bristowi* could indicate suboptimal experimental conditions, and hence results should be interpreted with care.*

We recognize that the tetrad experiment results are mainly driven by *M. bristowi*'s behavior as already mentioned in the results (line 231-232) but we now also mention it in the discussion (lines 401-402). This experiment would have benefited from more replicates, but the limited access to live males and virgin females for both subspecies was a limiting factor. Fisher's exact test used to assess assortative mating is specifically appropriate to small sample sizes. We recognize that the sampling size is not ideal, however it is still statistically testable.

*Regarding the wing manipulation experiment, *M. theodorus* does not show a preference when dummies with non-modified wings are presented and prefers non-modified dummies over modified dummies. This is acknowledged by the authors but not further discussed. Certainly, some control treatment for wing modification could have been added.*

The use of controls to consider the effect of wing modification and odor by the permanent marker were already mentioned in the methods (lines 636-639). Following your recommendation and comments from the other reviewers, we now mention the use of this control in the results (lines 278-283). We also address a potential issue that would have resulted in the rejection of these modified dummies by live males: we cannot be sure whether butterflies perceive these modifications as equivalent to natural coloration (lines 281-282). An additional control could have been used, adding black ink on the black dorsal parts of the pattern to assess its potential visual effect. The constraints on sampling unfortunately did not allow to add another treatment.

Overall, the fact that certain measurements only provide evidence for 1 of the 2 (sub)species (assortative mating, wing manipulation) or one sex of one of the species (bird visual systems) means overall interpretation and overgeneralization of the results to both allopatric or sympatric species should be done with care, and such nuances should ideally be discussed.

The aim of the authors, "to investigate the antagonistic effects of selective pressures generated by mate recognition and shared predation" has not been achieved, and the conclusions regarding this aim are not supported by the results. Nevertheless, the iridescence colour measurements are solid, and some of the behavioural experiments and chemical profile measurements seem to yield interesting results. The study would benefit from less overinterpretation of the results in the framework of predation and more careful consideration of methodological difficulties, statistical insecurities, and nuances in the results.

Overall, we would like to thank all reviewers for their thorough assessment of our work. We understand that the imbalance between mate choice data, visual model data and chemical data only gives us a partial assessment of species recognition in *Morpho* butterflies, thus requiring more precision in the interpretation and the discussion of our results. We made sure to add balanced interpretations in our discussion, by mentioning the lack of replicates for allopatric and sympatric populations (lines 391-392), and the lack of chemical characterization of allopatric species (lines 458-361, see previous comments) and by being more transparent on methodological limitations that we failed to convey in the first version of our manuscript. We brought nuance to our discussion and also discussed alternative hypotheses to predation to explain the convergence of iridescence found in sympatry.

Reviewing Editor Comments:

While all reviewers acknowledge the value of your data, they converge in their recommendations to tone down the evolutionary interpretations. Ideally, to test your main hypothesis, you would need several species pairs, or if only one, as in your case, replicated sympatric and allopatric sites for both species. Furthermore, your more specific hypotheses about convergence (vs. nondivergence), response to predators (vs. other environmental variables), and avoiding interspecific mating in sympatry (vs. not avoiding it in allopatry) would require appropriate alternative treatments/controls. We therefore recommend that you focus on those statements that you can support with your experiments and data, and introduce these statements in the introduction with reference to the appropriate literature.

Reviewer #1 (Recommendations for the authors):

(1) Line 25: This stated aim seems a bit off. The authors did not sensu stricto quantify 'how shared adaptive traits may shape genetic divergence' in this study. I suggest rewriting or deleting this whole sentence altogether. The study's aim is already clear in lines 29-34.

We deleted the mention of the characterization of genetic divergence, since this study did not focus on any genetic analysis.

(2) Line 34: The authors here state that they compared allopatric vs sympatric populations. This is strictly not true for M. Achilles. Further, the results after this sentence focus solely on divergence/convergence in sympatry, nothing at the intraspecific level and implications of the findings

We now mention that we tested allopatric vs. sympatric species of *M. helenor* only (lines 28-29). We also mention that the behavioral experiments were based on intraspecific comparisons, and discuss the implications of this result in the discussion.

(3) Line 35: 'convergence driven by predation': this is a strong statement and cannot be directly inferred from the present set of experiments. Consider toning it down.

We added nuance to this statement by rephrasing it “suggesting that predation may favor local resemblance” (lines 32-33)

(4) Line 36: Replace 'behavioral results' with 'behavioral experiments' or something similar.

Corrected

(5) Line 45-49: These opening statements need some citations.

We provided references for the first few lines, by citing terHorst et al 2018 (line 44) underlining the importance of species interactions in trait evolution, and Blomberg et al 2003 (line 45) showing that closely-related species tend to resemble each other by quantifying the phylogenetic signal of various traits.

(6) Line 83, 165: 'visual effect', not sure what the authors are referring to. Please rewrite.

We defined “visual effect” as the way wing color patterns could be perceived by predators or mates. We removed mentions of “visual effect” and directly used its definition instead.

(7) Line 105 onwards: This section of the introduction could benefit from more concise writing. The authors might consider reducing the number of specific examples and instead offering broader general statements, supported by citations from multiple studies.

We reduced the number of examples given in this paragraph and used general statements supported by multiple citations as examples. (lines 102-119).

(8) Line 108-110: This sentence seems to be redundant with the previous one.

We merged this sentence with the previous one to improve clarity. (lines 103-105)

(9) Line 140: 'with chemical defenses': include citations here.

We added citations of Joron et al 1999 and Merrill et al 2014, which document the evolution of convergent wing patterns (mimicry) in butterfly species with chemical-defenses.

(10) Line 149: *This is a bit of a stretch. Note that genetic divergence could be influenced by many other things, not only the processes that the authors examined.*

We agree with the reviewer that the study of the convergent vs. divergent evolution of visual cues is not enough to fully understand the mechanisms allowing genetic divergence between species. Because this paper does not focus on characterizing genetic divergence, we removed it from the manuscript to avoid oversimplification.

(11) Line 151: *Again. Here, the author's primary focus seems to be at an interspecific level. One is left to wonder about the need for comparisons at the intraspecific level in M.helenor and the implications. Please clarify*

In the end of the introduction (lines 146-157), we specifically highlighted the importance of intraspecific comparisons. While studying the effect of sympatry on the evolution of the iridescent color pattern, we use this intraspecific comparison as a baseline to account for convergence or divergence of iridescence in a sympatric interspecific pair of *Morpho*, because under neutral evolution two subspecies are expected to be more similar than two different species (this assumption has been clarified line 147-148). We also used intraspecific mate choice to test for the use of visual cues in mate recognition (experiment 1) and to test what type of signal could be perceived by *Morphos* (the iridescent coloration or the iridescent pattern, experiment 2 and 3). These results help contextualize the interspecific mate choice, focused on determining whether visual cues could also be used in species recognition. Since we show that iridescent coloration is important in mate recognition at the intraspecific scale, it helps understand why species recognition is low at the interspecific scale because of wing color convergence between *M. helenor* and *M. achilles*.

(12) Line 154: *'signals on mate preferences'.*

Corrected.

(13) Line 189: *'At the intraspecific level', maybe in the brackets include 'allopatric populations' just so the results are in a similar format as in the color contrast section below.*

We added details to make clearer that the intraspecific level is studied between allopatric *Morpho* populations (line 189).

(14) Line 189-192: *Please rearrange the figure (current B as A and vice versa) or present the results in order as in the figure (interspecific first and then intraspecific level).*

We rearranged Figure 3 so that the intraspecific comparison (allopatric population) appears as A and the interspecific level (sympatric population) appears as B, to follow the order of presentation in the main text.

(15) Line 232: *The motivation behind experiments 1, 2, and 3 is unclear. The authors have not made a strong point in the introduction about the need for these comparisons at an intraspecific level. Given that the authors are focused on divergence/convergence at an interspecific level, this set of experiments seems to be irrelevant to the present study. The implications of these findings are also not discussed.*

We added motivation to the use of experiment 1, 2, and 3 in the introduction (lines 151-154) by stating that those experiments were used to assess whether blue color could indeed be used as a mating cue in *Morpho helenor* (experiment 1) and to try to understand what part of

the visual signal is important in mate choice in *Morpho helenor*: the wing pattern (experiment 2) or the iridescent coloration (experiment 3). Although motivation for these experiments was not detailed in our manuscript, we already discussed the implications of the results of experiments 1, 2 and 3 in the discussion by stating that visual cues can take many forms and that considering both color AND pattern is important in understanding visual cues (lines 408-416). We carefully reworked this new version to make it more straightforward.

(16) Line 260: Insert 'wild-type' before model to ensure similar wording as in the previous section.

Corrected.

(17) Line 286: Insert 'sympatric' after mimetic.

Corrected.

(18) Line 307: Include a reference to the figures or table where these results are presented.

We now mention in the main text that the different proportions of beta-ocimene found between males *M. helenor* and *M. achilles* are shown in Table S2.

(19) Line 343: These inferences are speculative. Add a line here, something like 'although this warrants further research in this species'.

We detailed what additional experiments are needed lines 388-396.

(20) Line 357: The authors have not discussed their results on iridescence divergence in allopatric populations (line 190) and its implications.

We now made clear in the beginning of the discussion that the divergence of iridescence in allopatric populations is used as a baseline to test for convergent iridescence between species (lines 339-343).

(21) Line 361 onwards: This first paragraph is a bit confusing, as the results mainly focus on allopatry, while the title refers to sympatry.

To avoid confusion between the title and the content of the discussion, we divided the last part of the discussion into two different parts. As the first paragraph mainly focus on allopatry, we isolated it and titled it “Iridescent color patterns can be used as mate recognition cues in *M. helenor*” (line 498). The next paragraph of the discussion, focusing on the sympatric *Morpho* populations, has been titled “Evolution of visual and olfactory cues in mimetic sister-species living in sympatry” (line 418).

(21) Line 383: visual cues 'as' poor species.

Corrected.

(23) Line 405: Why females here and not males? This is again confusing since the authors tested for male mate choice in the main experiments. Some background information on sex-specific mate choice in the methods might help.

In this specific sentence, we talk about performing mate choice experiments to test for the discrimination of olfactory cues by females (and not males) because we found a high

divergence in the chemical compounds found on male genitalia. Although female chemical compounds could also be used as a cue by males in mate recognition, olfactive mate choice is often driven by female choice in butterflies. We recognize that this perspective does not line up with the mate choice presented in our results section which focused on male mate choice based on visual cues, because of ecological reasons (Morpho males tend to be attracted to bright blue colorations but not females) and technical reasons (in cages, females tend to hide away from the males or male dummies, and this behavior is not compatible with experiments involving flying around false males). In the discussion, we made sure to precise that the perspective we cite here is about testing the implications of divergence in male olfactory cues (line 454). We also added motivation to why we chose to investigate male (and not female) mate choice based on visual cues in the methods (lines 613-618) and in the results (219-223).

(24) Line 417: *This inference is speculative. Consider toning it down.*

We rewrote the sentence: “We find evidence of converging iridescent patterns in sympatry suggesting that predation could play a major role in the evolution of iridescence. Further work is nevertheless needed to directly test this hypothesis and establish the importance of evasive mimicry in Morpho” (lines 465-468).

(25) Line 429: *'Convergent trait evolution leads to mutualistic interactions enhancing coexistence'. Careful here. It is not very evident how convergent trait evolution (iridescence) is mutualistic in this case, as there is no experimental evidence for evasive mimicry yet. Consider rewording or toning this sentence down.*

We agree with the reviewer and removed this statement, only keeping the end of the sentence: “Altogether, this study addresses how convergence in one trait as a result of biotic interactions may alter selection on traits in other sensory modalities, resulting in a complex mosaic of biodiversity. (lines 479-481).

(26) Line 442: *Since the samples come from a breeding farm, I have a few questions. How are the authors sure about the location where the specimens were collected? How long have they been kept in captivity? Have they been subjected to any artificial selection? More details are needed here.*

Since *M. helenor bristowi* and *M. helenor theodorus* are only found in the wild in West and East Ecuador respectively, those *M. helenor* subspecies can only be collected in those two allopatric populations. Their phenotype is directly linked to their geographic repartition, this is how we made sure about their collect location. *M. h. theodorus* we used in this study were caught in East Ecuador in Tena, and *M. h. bristowi* were caught in West Ecuador in Pedro Vicente Madonado. We received pupae from the breeding farm, meaning that the Morpho used for the experiments were raised in captivity since their date of emergence. Upon emergence, they were transferred into cages for 4 to 5 days to wait for sexual maturity before performing the tetrad and mate choice experiments. This information was added to the method (lines 490-496).

(27) Line 476: *Include some citations supporting this statement.*

We now cite Bennett and Théry (2007), reviewing avian color vision, and Briscoe (2008), characterizing the sensitivity of the photoreceptors found in the eyes of butterflies. Both citations show that the 300-700nm range is seen by avian and butterfly visual systems.

(28) Line 480 onwards: *Please clarify if the analysis used only one value (mean?) per species, sex, angle of measurement, and locality or included data from multiple individuals.*

The analyses of both colorimetric variables and global iridescence were performed using iridescence data from multiple individuals (10 males and 10 females from *M. h. bristowi*, *M. h. theodorus*, *M. h. helenor* and *M. a. achilles*), for which we measured iridescence at 21 angles of illumination. Sampling size are mentioned lines 507, 515, 540-542.

(29) Line 510: *Is there a specific reason that authors did not investigate achromatic contrasts? Provide some justification here. Or include the results of achromatic contrasts in the supplement.*

We added the achromatic results in the supplement and in the results (lines 200-204). For both the avian visual model and the Morpho visual model, the confidence intervals always overlapped with the JND threshold, showing that neither birds nor butterflies could theoretically discriminate the wing reflectance brightness in allopatric and sympatric populations.

(30) Line 552 onwards: *I may have missed it. It is not entirely clear why the authors focused on male mate choice rather than female preference for visual cues. The authors should explicitly justify this choice and cite previous studies demonstrating that male mate choice, rather than female preference, is important in this species. This should be stated in the results section as well.*

We added a paragraph in the method (lines 613-618) to describe the ecological and technical reasons leading to testing only male mate choice using visual cues (also see our response to recommendation #23).

(31) Line 537 onwards: *What was the criterion used to score that mating had occurred? Why first mating and not how long they were mating? Please add these details.*

We stopped the experiment as soon as a male/female pair was formed by joining their genitalia (we added this information in the method lines 599-600). Since the tetrad experiment involves the interaction of two males and two females from different subspecies, we considered that mate choice happened before the formation of any couple, and is not necessarily dependent on how long they mate by observing their mating behavior. For instance, we witnessed avoidance behaviors from females that systematically hide their genitalia and refused to join their abdomen to some males, while being very 'open' to others (but did not quantify it).

(32) Line 571: *The authors used a black permanent marker to modify wing patterns but did not validate whether butterflies perceive these modifications as equivalent to natural coloration. It is possible that the alterations introduced unintended visual cues and may explain why most males rejected the dummies (line 267). The authors should acknowledge this limitation here.*

We now acknowledge this limitation in the method (lines 638-639) and in the results section (lines 278-283).

(33) Line 591: *Insert 'above' after protocol.*

Corrected.

(34) Line 605: *If the authors included random effects in their model, then it should be generalized linear mixed model (GLMM) and not GLM as they wrote.*

We indeed included a random effect in our model accounting for male ID and trial number, we thus replaced “GLM” by “GLMM” in the manuscript.

(35) Line 615: This set of analyses does not seem to account for pseudo-replication, as the data were recorded from the same male more than once (Line 583). Please clarify and redo the analysis with the GLMM framework

We run new analyses using the GLMM framework: we used a binomial GLMM to test whether individuals preferentially interacted with dummy 1 vs. dummy 2 while accounting for pseudoreplication. The previously detected tendencies hold true with these new analyses, except for the visual mate discrimination of *M. achilles*: we now find statistical evidence that *M. achilles* tend to approach more their conspecifics during the mate choice experiment, although the signal is weak (line 297-307). Indeed, while we previously concluded that both species in sympatry (*M. helenor* and *M. achilles*) could not discriminate their conspecific mates, we now emphasize that *M. achilles* is somewhat sensitive to some visual signals. However, its estimated probability of approaching a conspecific is only 0.54, which is low compared to the estimated probability of approaching (0.61) or touching (0.84) a con-subspecific for *M. bristowi*. We thus concluded that even though some visual cues could be relevant for mate recognition, they are less reliable for male choice in sympatric populations where color patterns are more convergent, compared to allopatric populations. We thus updated Figure 4 and Figure S8 and S9, which are now picturing the probability of approaching or touching a conspecific or con-subspecific with the updated p-values retrieved from the GLMM analyses. We also updated the results (line 297-307) and the discussion (lines 430-438) to bring nuance to our previous results.

*(36) Line 963: Figure 3D. Is there a particular reason for comparing allopatric populations only within Ecuador rather than between Ecuador and French Guiana for *M. helenor*? Please clarify.*

We aimed at comparing the putative discrimination of blue coloration using visual models vs. what the butterflies actually discriminate using mate choice experiments. Since we only performed mate choice experiments involving *M. h. bristowi* x *M. h. theodorus* (allopatric populations within Ecuador) and *M. h. helenor* x *M. a. achilles* (sympatric population from Ecuador), we only looked at those comparisons using visual models. We added this precision lines (559-560).

(37) Line 980: Are these predicted probabilities or just mean proportions as written in line 614? Then the label should be changed to 'Proportion of approaches' or something similar.

Following our answer to recommendation #35, the points now represent the probability of touching a conspecific in the graph for each male, for every trial of every male tested. We corrected the legend of the figure.

Reviewer #2 (Recommendations for the authors):

(1) Line 25: "...therefore facilitating co-existence in sympathy".

Corrected.

(2) Line 28: "contrasting" instead of contrasted.

Corrected.

| (3) Line 33: begin a new sentence at the colon.

Corrected.

| (4) Line 49: the phrase "habitat filtering" is unclear and should perhaps be defined or qualified.

We replaced "habitat filtering" by its definition and cited Keddy (1992), describing the community assembly rules and defining habitat filtering (line 46)

| (5) Line 52: remove "even".

Corrected.

| (6) Line 53: divergent suites may also result because traits are often constrained by genetic architecture (multivariate genetic covariances). This is discussed at length and specifically in relation to ornamental coloration by Kemp et al. 2023

We rewrote the introduction and focused on only reviewing the ecological interactions promoting trait divergence in sympatric species, and did not mention genetics in this paper.

| (7) Line 87: (and throughout) refer to "colouration" or "colour pattern" rather than "colourations".

Corrected.

| (8) Line 151: Remove "To do so,".

Corrected.

| (9) Line 191: I would like to see the degrees of freedom for this test.

We added the F-statistic=2.09 and the degrees of freedom df=1 of this test, and for all the following tests.

| (10) Line 201: (and throughout) replace "on" with "of".

Corrected.

| (11) Line 205: modelling the visual properties of the wings allows one to infer what is theoretically visible/distinguishable. The modelling is useful but not necessarily definitive of vision/behaviour per se under different conditions in the wild. I therefore think it is appropriate to phrase the wording around the modelling approach more carefully. Perhaps refer to "theoretical" or "inferred" discriminability, or state (e.g.) that species should/should not be capable of perceiving differences based on the modelling data. You do this well in your wording of lines 207-209. This need not apply in the discussion because you're then dealing with the combination of modelling results and behaviour (mating trials).

We agree with the reviewer that visual modelling only allows to infer what is theoretically discriminated by the butterflies, and that the wording of our sentence is confusing. We therefore modified the sentence to account for those precisions: "Morpho butterflies and predators can theoretically visually perceive the difference in the blue coloration between

different subspecies of *M. helenor*..... using both bird and Morpho visual models” (line 206-209).

(12) Line 222: *Either the chi-square test or Fisher's exact test should be sufficient (why report both?)*

Chi-square test relies on large-sample assumptions (expected counts > 5) whereas Fischer's exact test does not and is valid even with small or unbalanced sample sizes. Since the *M. bristowi* female/*M. h. theodorus* male pairing only occurred 3 times, we do not meet the primary assumptions to apply a Chi-square test, although it is significant. We used a Fischer's test to confirm the results. Using both and finding that both tests are significant shows that the results are robust, although they may appear redundant. To simplify, we remove the results of the Chi-square test and only keep the Fisher's test in the methodology and the results.

(13) Line 224 (and throughout): *Degrees of freedom should be provided for statistical tests.*

We reported the statistic value and the degrees of freedom for all mentions of the statistical tests in the main text, except for the Fischer test which does not rely on an asymptotic distribution like the Chi-squared distribution as it is an exact test.

(14) Lines 266-267: *This sentence has interest, but it is rather vague at present. Wouldn't your controls account for the effect of manipulation? This could be explained further.*

During our mate choice experiments, all Morpho female dummies used for the experiments were painted with black markers, either on their dorsal blue band to modify their blue iridescent phenotype, or on their ventral side, thus controlling for the effect of manipulation. However, we cannot rule out that the modification of the dorsal blue iridescence could have had a “repulsive” effect for males for several reasons. For example, depending on the visual discrimination of darker colors by Morphos, the painted black band could have a slightly different color compared to the dark “brown” usually surrounding their blue iridescent patterns. We now explain this in the results (lines 278-283) and in the methodology (lines 638-639)

(15) Line 316: *I'm not certain that the similarity is best described as "striking", given a P-value of 0.084 for this contrast*

We agree with the reviewer and removed this adjective for this line.

(16) Lines 387-390: *This sentence is puzzling because, theoretically speaking, we should expect selection on visual preference to be heightened (not relaxed) in sympatry if colouration is included among the traits used in mate selection. I'm not certain I have understood the meaning here.*

We would like to thank the reviewer for pointing out this typo. If shared predatory pressures favors convergent evolution of color pattern, then the visual signals become less reliable for species recognition. As a result, sexual selection on visual preference is heightened and becomes stronger, favoring the evolution of alternative cues used to discriminate conspecific mates. We changed the sentence and now write “the convergent evolution of iridescent wing patterns... may have negatively impact visual discrimination and favored the evolution of divergent olfactory cues” (lines 457-458).

(17) Line 529: Mating experiments. Given that these are quite large butterflies, I wondered whether a 3x3x2m cage would be sufficient in size to allow the expression of male courtship. A brief description of the courtship behaviour in these species or Morphos generally would be a useful addition to the paper.

A cage this size was enough for the males to express a flight behavior similar to what can be seen in nature, while also being able to see the females (live females or dummies). We tried to perform mate experiments in a larger cage (7m x 5m x 3m) but the trials were not conclusive because male did not find the dummies depending on where they were flying in the cage. A 3mx3mx2m cage is a good compromise maximizing interactions while still allowing enough space to fly. We now describe Morpho male behavior and female behavior in the methods (lines 613-618).

(18) Line 546: Why are both tests needed (chi-square AND Fisher's exact)?

Similarly to our answer on recommendations #12, we used both tests to show robustness in the statistical results. We only kept the Fisher's test results to simplify the results.

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